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A meta-analysis and meta-regression to compare the production performance, feed efficiency, and milk N efficiency in Jersey versus Holstein cows

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ABSTRACT

A meta-analysis and a meta-regression were conducted to compare DMI, production performance, feed efficiency (FE), and milk N efficiency (MNE; milk N yield/N intake) of Holstein and Jersey cows. Only experiments in which purebred Holsteins and Jerseys enrolled in the same study, under similar experimental conditions, were used to build the datasets (n = 30 peer-reviewed articles). The weighted raw mean difference (WMD) between breeds was computed for each comparison, and weights were assigned based on the inverse of the variance. A meta-regression was run to explore the effect of continuous (year of publication, DIM, and dietary concentrations of NDF, CP, and forage) and categorical (forage type: corn silage, alfalfa/grass-clover mix, and other forage sources ["others"]) covariates on the WMD of different variables. Jerseys had lower DMI (−4.15 kg/d), milk yield (−10.8 kg/d), yields of milk fat (−61.6 g/d) and milk true protein (−211 g/d), and ECM yield (−5.11 kg/d) than Holsteins. In contrast, concentrations of milk fat (+16.1 g/kg) and milk true protein (+6.98 g/kg), DMI (% of BW [+0.44%] and % of metabolic BW [MBW = $BW^{0.75}$; +0.64%], and ECM yield (% of BW [+0.64%] and % of MBW [+1.63%]) were greater in Jersey than Holstein cows. Whereas FE (milk yield/DMI) was lower (−0.26 kg/kg) in Jerseys than Holstein cows, breed did not affect FE (ECM yield/DMI) and MNE. Year of publication widened the WMD of milk fat and milk true protein concentrations and ECM yield (% of MBW), and it narrowed that of DMI (kg/d), milk true protein yield, and N intake, favoring Jerseys over Holsteins in recent publications. In addition, DIM narrowed the gap between breeds for the variables yields of milk, ECM yield (kg/d), milk fat, and milk N yield, and widened the gap for DMI (% BW and % of MBW) and ECM yield (% of BW and % of MBW), all in favor of Jerseys as lactation progressed. Dietary concentrations of NDF, CP, and forage collectively affected the WMD of BW, yields of milk and milk lactose, milk true protein and milk lactose contents, ECM yield (% of BW and % of MBW), FE (milk yield/DMI), and N intake, but did not consistently benefit Holsteins or Jerseys. Corn silage affected all WMD, except milk fat yield, ECM yield (% of BW and % of MBW), FE (ECM yield/DMI), and MNE. Relative to corn silage, alfalfa-grass/clover mix diets affected the WMD of DMI (kg/d and % of MBW), milk yield, milk fat and milk true protein yields, ECM yield (kg/d), N intake, and milk N yield, favoring Holsteins over Jerseys. Compared with corn silage, the forage type others affected the WMD of milk fat and milk true protein concentrations, DMI (% of BW), and ECM yield (% of BW and % of MBW), benefiting Jersey versus Holstein cows. In conclusion, the WMD of FE (ECM yield/DMI) and MNE were similar between breeds and the continuous (year of publication and DIM) and categorical (forage type) covariates largely influenced the WMD of several variables.

Key words: dairy breed, feed intake, milk performance, nitrogen utilization

INTRODUCTION

The US dairy herd population consists of approximately 9.4 million cows (USDA-ERS, 2024), with approximately 79.7% Holsteins, 8.2% Jerseys, and 12.1% other breeds (CDCB, 2021). Olthof et al. (2023), evaluating the profitability of 3 commercial dairies from Michigan, concluded that reduced expenses attributed to the smaller body size and lower feed requirement of Jerseys did not fully compensate for the revenue lost when compared with Holsteins. In fact, Jersey

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

cows were less profitable than Holstein cows due to their lower milk yield (**MY**) and production of milk components (Olthof et al., 2023). However, life-cycle assessment studies revealed that relative to Holsteins, milk and cheese produced from Jerseys resulted in less use of natural resources and a decreased C footprint (Capper and Cady, 2012; Uddin et al., 2021; Børsting et al., 2023). Therefore, there are potential economic and environmental tradeoffs between these 2 breeds, indicating that selection of Jersey and Holstein cows would depend on individual farm goals, production system (e.g., confinement, grazing), and market conditions, including milk pay prices, milk quality bonus and premium structures, and feed costs.

Aikman et al. (2008) showed that compared with Holsteins, Jerseys spent more time eating and ruminating per unit of ingested feed. They also demonstrated that feed passed through the gastrointestinal tract of Jersey cows faster than that of Holstein cows. These differences in chewing behavior and digesta passage rate reported by Aikman et al. (2008) may interact to ultimately affect feed efficiency (**FE**) between these 2 breeds. For instance, Uddin et al. (2020) reported lower FE, calculated as MY divided by DMI (**FE_{MY}**), in Jersey versus Holstein cows fed diets in which alfalfa silage NDF partially replaced corn silage NDF at 2 levels of forage NDF. Furthermore, Olijhoek et al. (2022) showed that FE, computed as ECM yield divided by DMI (**FE_{ECM}**), was greater in Jerseys than Holsteins in diets with concentrate levels varying from 49% to 91% (DM basis). In contrast to FE_{MY} and FE_{ECM}, milk N efficiency (**MNE** = [milk N/N intake]) was not affected by breed even in diets with different dietary levels of CP and NDF, forage:concentrate (**F:C**) ratios, and silage sources (Kauffman and St-Pierre, 2001; Uddin et al., 2020; Børsting et al., 2023). Although information from all the individual studies cited in this section are helpful to evaluate the effect of breed on MY, FE, and MNE under specific dietary and management conditions, pooling data from multiple experiments through a meta-analysis and meta-regression approach is a more robust methodology for making decisions and drawing conclusions regarding the production performance and nutrient use efficiency in Holstein versus Jersey cows. We are not aware of any studies to date that have compared Holsteins versus Jerseys by integrating meta-analytical methods with meta-regression and subgroup analyses to identify animal and nutritional factors contributing to breed variability in DMI, MY, milk composition, and nutrient use efficiency, assessed through FE and MNE.

The objective of this study was to compare production performance, FE, and MNE in Holstein versus Jersey cows through meta-analysis and meta-regression. Our central hypothesis was that FE_{ECM} would be greater in Jerseys than Holsteins, with MNE not affected by breed.

MATERIALS AND METHODS

Literature Search and Inclusion Criteria

A comprehensive literature search using the databases Web of Science (https://www.webofscience.com), ScienceDirect (https://www.sciencedirect.com), and CABI Digital Library (https://www.cabidigitallibrary.org) was conducted to identify publications comparing DMI, MY, milk composition, FE, and MNE of purebred Holstein versus Jersey cows. The literature search covered titles and abstracts of indexed articles using the following combination of key words: "dairy cows" OR "dairy cow" OR "*Bos taurus*," "Jersey" AND "Holstein," "dry matter intake" OR "DMI" AND "feed efficiency" OR "FE," and "milk production" OR "milk yield" OR "milk." Each combination of key words was entered in individual rows and searched simultaneously using the advanced search tools of Web of Science, ScienceDirect, and CABI Digital Library. Articles were considered candidates for the meta-analysis only when the following inclusion criteria were met: (1) presence of purebred Holstein and Jersey cows in the same study under similar experimental conditions; (2) reported individual DMI for each breed, including grazing studies in which markers were used to estimate herbage DMI; (3) reported MY and milk composition per individual breed; (4) reported SEM, standard error of difference (**SED**), or SD for estimating the variance of the variables of interest; and (5) published in English between 1974 and 2024.

The initial literature search yielded 785, 105, and 20 studies from the Web of Science, ScienceDirect, and CABI Digital Library, respectively, resulting in 910 potential studies for building the datasets. Review articles (n = 18), supplemental data (n = 9), dissertations and theses (n = 8), and case studies (n = 3) were excluded during our initial screening. The remaining candidate papers (n = 872) were further screened for duplications and suitability for inclusion according to the following steps: step (1) abstracts were read to confirm that the studies used purebred Holstein and Jersey cows; step (2) the materials and methods sections were examined to exclude experiments in which breeds were not treated under the same experimental conditions (e.g., facilities, dietary treatments, lactation stage); and step (3) the results or results and discussion sections were reviewed to ensure that authors reported a measure of dispersion for the variables of interest and individual DMI. A total of 846 candidate papers were excluded based on steps 1 through 3, resulting in 26 peer-reviewed articles included in the initial phase of papers selection. Four additional publications were excluded during further screening due to the absence of a measure of dispersion for the individual breed effect, resulting in 22 papers. In the second phase, all 22 papers were read thoroughly, including their reference lists, resulting in the inclusion of 10 peer-reviewed articles not identified in our initial literature search, possibly because the terms Holstein and Jersey were not reported in the titles, abstracts, and indexing key words. In addition, some publications were only available through scanned documents rather than text-based PDF files, likely compromising the proper indexing of key words and searchability. During the final screening, 1 article (Muller and Botha, 2009) was excluded because DMI was not measured individually despite authors conducting the statistical analyses using cow as the experimental unit. A second article (Munksgaard et al., 2020) was also excluded due to unclear information on how total DMI was calculated, with concentrate allowance varying from 2 to 10 kg/d. After excluding these 2 papers, 30 peer-reviewed publications were included in the final datasets as summarized in Figure 1. The complete set of references used in the present meta-analysis is provided in Supplemental File S1 (see Notes; Almeida et al., 2026a).

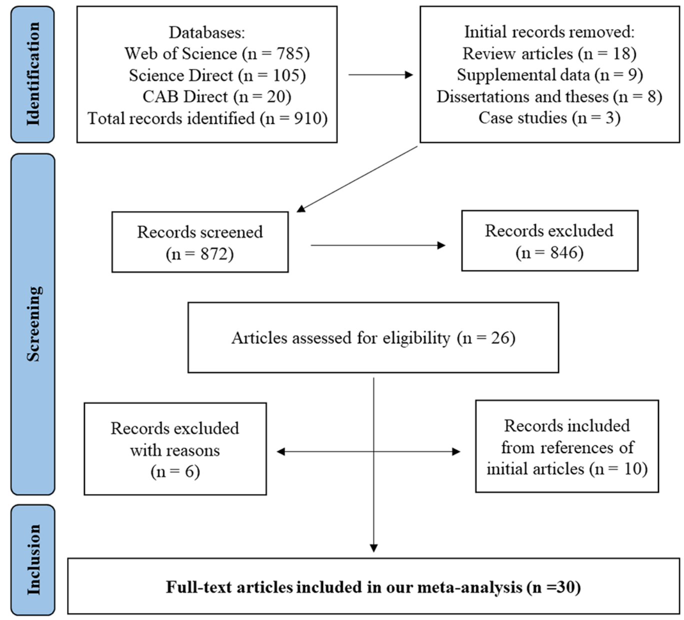

Figure 1. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram (Moher et al., 2009) of the systematic review from the initial search, screening, and final selection of studies included in the meta-analysis and meta-regression comparing DMI, production performance, FE, and MNE of Holstein versus Jersey cows.

Data Extraction and Calculations

Data showing the number of cows, DIM, dietary treatments, feeding management, forage type, and F:C ratio were extracted from the materials and methods section and tables of the selected studies. Response variables (i.e., BW, DMI, MY, concentrations and yields of milk components, apparent total-tract digestibility of NDF [**NDFD**]) from which we were able to extract information, such as the number of comparisons (i.e., observations) per breed, LSM, and measure of dispersion, were classified as "observed" variables. Response variables (e.g., ECM yield, FE, and MNE) not consistently reported across the selected studies were recalculated using mathematical equations and classified as "estimated" variables. Yield of ECM was calculated according to Sjaunja et al. (1990) as follows: ECM yield (kg/d) = [0.25 × MY (kg/d)] + [12.2 × milk fat yield (kg/d)] + [7.7 × milk CP yield (kg/d)]. This equation, without milk lactose yield, was used because it allowed the inclusion of a larger number of observations from the selected studies. When not specified, milk samples analyzed by midinfrared spectrophotometry were assumed to represent milk true protein concentration. For studies reporting milk CP concentration, milk true protein was calculated by subtracting 5% from milk CP to account for the NPN fraction (DePeters and Ferguson, 1992). Milk true protein yield was converted to milk CP yield to calculate ECM yield when needed. Feed efficiency was calculated by dividing MY by DMI (i.e., FE_{MY}) and ECM yield by DMI (i.e., FE_{ECM}). When not reported, CP intake was calculated by multiplying DMI by the corresponding treatment CP concentration and converted to N intake using the 6.25 conversion factor. Milk N yield was obtained by dividing milk true protein yield by 6.38, and MNE was calculated by dividing milk N yield by N intake and multiplying by 100.

The year of publication of selected studies ranged from 1978 to 2023, with 2008 calculated as the median year, which was used as the threshold in the subgroup analyses. Days in milk was calculated as the average between initial and final DIM (last day of the experiment). Dietary concentrations (% of DM) of NDF and CP were back-calculated from reported NDF and CP intakes when NDF and CP contents were not reported in the studies. If diet composition, nutrient intake, and forage level were unavailable, the missing data (n = 12 [NDF, % of DM], n = 1 [CP, % of DM], n = 1 [forage, % of DM]) were replaced by the average concentrations of their respective values reported in the selected papers. This was done to maximize the number of breed comparisons in the meta-regression analysis. Forage type was divided into 3 categories, with the first 2 categories defined based on the forage type that accounted for at least 55% of the forage component in the diet DM. The third category included diets with ≥2 forage sources at equal or similar inclusion levels in the diet DM, but below the minimum 55% forage DM threshold, and forage sources with a low number of observations (i.e., freshly-cut forage or grazed herbage; n = 4). Forage types were classified as follows: (1) corn silage–based diets ("corn silage"; n = 26 observations), (2) alfalfa (*Medicago sativa* L.) hay- or alfalfa silage–based diets and grass-clover mix silage (“alfalfa/grass-clover mix”; n = 14 observations), and (3) forage sources that did not fit in any of the other 2 categories already described (“others”; n = 35 observations).

For estimated variables, the measure of dispersion (i.e., SD, SED, or SEM) of closely related variables was used to account for the variance between breeds. For instance, the measure of dispersion of DMI (kg/d) was used in the analyses of DMI (% of BW and % of metabolic BW [**MBW** = $BW^{0.75}$]), N intake, FE_{MY}, FE_{ECM}, and MNE, and that of MY (kg/d) for analyses of ECM yield (kg/d, % of BW, and % of MBW). Furthermore, in the analysis of milk N yield, 2 approaches were adopted to propagate the error: (1) the measure of dispersion of milk true protein yield divided by 6.38 was used when reported, and (2) the measure of dispersion of MY and milk protein concentration were used for missing values as follows: SEM of milk N yield = estimated milk true protein yield × $[(\text{SEM of MY/MY})^2 + (\text{SEM of milk true protein concentration/milk true protein concentration})^2]^{0.5}$. All data were compiled into Excel spreadsheets (version 2311, Microsoft Corp.) and stratified by study and breed. The final dataset used for the statistical analyses, including calculations, trimmed propagation of error, subgroups, and outliers, can be found in Supplemental File S2 (see Notes; Almeida et al., 2026b).

Statistical Analyses

Effect Size and Weighting. The effect of breed (Jersey vs. Holstein cows) on DMI, production performance, FE, MNE, N intake, and NDFD was evaluated using weighted raw mean differences (**WMD**). The WMD was weighted by the inverse of the variance within a random-effects model using the REML method for estimating variance components (Viechtbauer, 2010). The SD needed to estimate WMD was either obtained from the reported SD of variables of interest or calculated from the reported SEM as follows (Vesterinen et al., 2014):

$$\text{SD} = \text{SEM} \times \sqrt{n},$$

where n = number of experimental units in individual experiments. Studies were grouped based on the type of statistical model used (i.e., MIXED, GLM) to account for potential differences in model error structures. Scaling of SEM within each statistical procedure was adapted from St-Pierre (2001) and Roman-Garcia et al. (2016). First, reported SEM values for individual variables were divided by their respective averaged SEM and trimmed using a threshold of 0.35. The SEM ratios averaged 1.0, and each ratio was then multiplied by the reported SEM to obtain the normalized SEM, which was used to calculate SD. Each variance comparison between Jersey and Holstein cows was calculated using the square of the pooled SD (Vesterinen et al., 2014). Forest plots were created using STATA software (version 19.5; StataCorp, 2025) to visually represent the findings by showing the mean size effect for each observation, including intra-study comparisons, and indicating the relative weight of each study to the overall size effect estimate with the respective 95% CI for the size effect.

Heterogeneity. Variation between breeds was assessed using the χ^2 (Q) test of heterogeneity and I^2 statistic (Higgins et al., 2003). Note that I^2 represents the proportion of the true variance in treatment effect relative to the total variance observed for a given treatment (Borenstein et al., 2017; Lean et al., 2018). Negative I^2 were assigned a value of zero, with values of <25%, 25% to 50%, and >50% denoted as low, moderate, and high heterogeneity, respectively (Higgins et al., 2003).

Meta-Regression and Subgrouping. A meta-regression analysis was conducted to identify the effect of continuous and categorical covariates on WMD. We used a mixed-effects model with WMD as the dependent variable and the REML method for estimating variance components (Viechtbauer, 2010) to fit the continuous covariates year of publication (1978 to 2023), DIM, and dietary concentrations (% of DM) of NDF, CP, and forage as described below:

$$\Theta_i = \beta_0 + \beta_j x_{ij} \ldots + \beta_p x_{ip} + \mu_i,$$

where Θ_i = WMD of study i, β_0 = intercept, β_j = coefficient for the covariate j, x_{ij} = value of the covariate j (1, 2, ... [p]) in study i, and μ_i ~N (0, τ^2) = between-study random effect (heterogeneity). A correlation analysis was conducted, yielding a maximum correlation coefficient (r) across covariates of 0.47. Reduced models were then built using a backward elimination approach, iteratively excluding covariates with $P > 0.10$. A second mixed-effects meta-regression model was fitted for the categorical covariate forage type, which was split into 3 categories as mentioned earlier, using WMD as the dependent variable. Corn silage was used as the reference group in the model (i.e., the intercept) because it was the easiest to classify, with the forage types alfalfa/grass-clover mix and others fitted as the slopes according to the statistical equation described below:

$$\Theta_i = \beta_0(\text{corn silage}) + \beta_j x_{ij} \ldots + \mu_i,$$

where Θ_i = WMD of study i, β_0 = intercept using corn silage as the reference group, β_j = coefficient for the covariate j (i.e., the difference between the true coefficient

Table 1. Descriptive statistics of observed variables in Holstein and Jersey cows[1]

Item	n (Study)	n (Comparison)	Mean	±SD	Minimum	Maximum
Holstein						
DMI, kg/d	30	75	21.0	2.24	16.1	25.3
BW, kg	18	45	617	57.6	498	718
MY, kg/d	29	72	31.8	6.09	16.9	44.9
Milk fat, g/d	17	44	1,161	225	660	1,619
Milk true protein, g/d	17	44	1,038	162	602	1,321
Milk lactose, g/d	8	19	1,715	103	1,570	1,981
Milk fat, g/kg	27	71	37.2	5.69	24.5	56.7
Milk true protein,[2] g/kg	27	71	32.5	2.98	27.4	37.9
Milk lactose, g/kg	16	40	48.5	1.45	43.5	50.3
NDFD, % of NDF intake	8	14	55.8	11.1	41.4	78.6
Jersey						
DMI, kg/d	30	75	16.8	2.23	12.3	22.8
BW, kg	18	45	440	35.6	369	541
MY, kg/d	29	72	21.1	3.79	12.8	29.7
Milk fat, g/d	17	44	1,084	238	660	1,550
Milk true protein, g/d	17	44	824	182	518	1,188
Milk lactose, g/d	8	19	1,177	136	940	1,430
Milk fat, g/kg	27	71	53.3	8.70	37.3	76.9
Milk true protein,[2] g/kg	27	71	39.2	4.47	32.6	47.7
Milk lactose, g/kg	16	40	47.5	1.52	42.8	50.6
NDFD, % of NDF intake	8	14	57.7	12.2	42.7	81.0

[1]Observed variables were obtained from breeds LSM presented in the selected studies.

[2]Milk true protein concentration was calculated by subtracting 5% from milk CP to account for NPN (DePeters and Ferguson, 1992) when not reported in the selected publications.

of the covariate and the intercept), x_{ij} = category of the covariate j (alfalfa/grass-clover mix or others) in study i, and $\mu_i \sim N(0, \tau^2)$ = between-study random effect (heterogeneity). Tests of the null hypothesis for the covariate coefficients were obtained from the multiparameter Wald test (Harbord and Higgins, 2008). The adjusted R^2 was calculated by comparing the estimated between-study variance including (σ^2) or excluding $\left(\sigma_0^2\right)$ covariates from the model as follows: adjusted $R^2\ (\%) = \left(\sigma_0^2 - \sigma^2\right) / \sigma_0^2$. The adjusted R^2 represents the proportion of between-study variance (i.e., heterogeneity) explained by the covariates (Harbord and Higgins, 2008; Viechtbauer and Cheung, 2010).

A subgroup analysis was conducted to further explore breed differences when the effect year of publication, DIM, or dietary forage level (DM basis) on the WMD of a given variable was significant ($P \le 0.05$). Subgroups were defined as follows: (1) year of publication ≤2008 and >2008, (2) cows at ≤100 DIM and >100 DIM, and (3) diets with ≤55% and >55% of forage in the total diet DM. The thresholds for year of publication, DIM, and forage level were set to investigate potential differences between breeds in genetic progress, lactation stage, and F:C ratio, respectively. A test of moderator (Viechtbauer, 2010) was used to determine whether groups (e.g., ≤100 DIM vs. >100 DIM) were different ($P \le 0.05$) from each other.

Publication Bias and Outlier Analysis. Publication bias was assessed using funnel plots (Light and Pillemer, 1984), with funnel plot asymmetry tested using the Egger's regression method between WMD and SE for both observed and estimated variables (Egger et al., 1997). Outliers were identified by evaluating their influence on the comparisons between breeds using Cook's distance (Cook, 1977).

Statistical Packages. The "metafor" package (Viechtbauer, 2010) from R Studio (version 4.0; R Core Team, 2020) was used for overall WMD, meta-regression, subgroup, publication bias, and outlier analyses. Significance was declared at $P \le 0.05$ and tendencies at $0.05 < P \le 0.10$. A generic R script used for the statistical analyses is provided in Supplemental File S3 (see Notes; Almeida et al., 2026c).

RESULTS

Data from 30 peer-reviewed articles with a total of 1,479 cows (806 Holsteins and 673 Jerseys) were included in the final datasets for statistical analyses comparing DMI, production performance, FE, MNE, and NDFD in Holstein versus Jersey cows. Descriptive statistics for both observed and estimated parameters are presented in Tables 1 and 2, respectively. Descriptive statistics for covariates used in the meta-regression and subgroups analyses are shown in Table 3. Forest plots with the WMD of DMI stratified by year of publication and BW stratified by DIM are shown in Figures 2 and 3, respectively.

Table 2. Descriptive statistics of estimated variables in Holstein and Jersey cows[1,2]

Item	n (Study)	n (Comparison)	Mean	±SD	Minimum	Maximum
Holstein						
DMI, % of BW	20	51	3.43	0.41	2.49	4.39
DMI, % of MBW	20	51	17.1	1.83	12.6	20.9
ECM yield, kg/d	26	68	30.9	5.61	17.2	41.9
ECM yield, % of BW	17	45	5.02	0.85	3.40	6.41
ECM yield, % of MBW	17	45	25.1	4.19	16.2	32.3
FE_{MY}, kg/kg	29	72	1.51	0.23	1.01	2.00
FE_{ECM}, kg/kg	26	68	1.45	0.20	1.00	1.91
N intake, g/d	29	74	583	80.1	409	774
Milk N yield, g/d	26	68	163	31.1	91.0	220
MNE, % of N intake	26	68	27.9	5.61	15.3	38.8
Jersey						
DMI, % of BW	20	51	3.83	0.51	2.60	4.97
DMI, % of MBW	20	51	17.6	2.29	12.2	22.9
ECM yield, kg/d	26	68	25.7	4.99	15.9	34.7
ECM yield, % of BW	17	45	5.87	0.92	4.19	7.44
ECM yield, % of MBW	17	45	26.9	4.47	18.7	34.5
FE_{MY}, kg/kg	29	72	1.26	0.18	0.88	1.71
FE_{ECM}, kg/kg	26	68	1.51	0.21	1.09	1.94
N intake, g/d	29	74	467	73.3	328	638
Milk N yield, g/d	26	68	130	28.0	79.5	186
MNE, % of N intake	26	68	27.7	5.07	14.8	36.6

[1]Estimated variables were calculated based on data obtained from the selected studies.

[2]Initial BW was used to calculate DMI and ECM yield, as proportions of BW and MBW, when not reported in the selected publications.

Weighted Raw Mean Differences of Observed Variables

Breed type affected ($P \leq 0.01$) the WMD of all variables reported in Table 4. Compared with Holsteins, Jersey cows had lower DMI (WMD = −4.15 kg/d), BW (WMD = −169 kg), MY (WMD = −10.8 kg/d), milk fat yield (WMD = −61.6 g/d), milk true protein yield (WMD = −211 g/d), and yield (WMD = −584 g/d) and concentration (WMD = −0.91 g/kg) of milk lactose. In contrast, Jerseys had greater concentrations of milk fat (WMD = +16.1 g/kg) and milk true protein (WMD = +6.98 g/kg), and NDFD (WMD = +1.88%) than Holsteins. High heterogeneity ($I^2 \geq 75.5\%$; $P < 0.001$) was observed for all variables presented in Table 4. In addition, publication bias, based on the funnel plot asymmetry test, was detected ($P \leq 0.03$) for DMI, MY, and yields of milk fat and milk true protein, with tendencies ($P \leq 0.08$) observed for milk fat concentration and NDFD. The funnel plots for selected observed variables are presented in Figure 4A–F.

Weighted Raw Mean Differences of Estimated Variables

Breed type affected ($P \leq 0.003$) the WMD of all estimated variables, except for FE_{ECM} ($P = 0.56$) and MNE ($P = 0.33$), as shown in Table 5. Jersey cows had greater DMI (% of BW and % of MBW; WMD = +0.44% and +0.64%, respectively) and ECM yield (% of BW and % of MBW; WMD = +0.64% and +1.63%, respectively) than Holstein cows. Contrarily, ECM yield (WMD = −5.11

Table 3. Descriptive statistics of covariates used in the meta-regression and subgroup analyses

Item	n (Comparison)	Mean	±SD	Minimum	Maximum
DIM[1]	75	137	48.4	9	272
≤100 DIM	19	75.3	22.5	9	96
>100 DIM	56	158	34.8	108	272
NDF,[2] % of DM	63	33.0	4.46	24.8	46.6
CP,[2] % of DM	74	17.3	1.82	13.9	23.9
Forage, % of DM	74	55.4	15.6	8.79	100
≤55% of forage DM	43	46.3	10.0	8.79	55.0
>55% of forage DM	31	68.0	12.8	58.0	100

[1]DIM was calculated as the average of the initial and final DIM (i.e., last day of the experiment).

[2]The dietary concentrations of NDF and CP were back-calculated from NDF and CP intake when not reported in the selected publications.

kg/d), FE_{MY} (WMD = −0.26 kg/kg), N intake (WMD = −115 g/d), and milk N yield (WMD = −32.2 g/d) were lower in Jerseys compared with Holsteins. High heterogeneity ($I^2 \geq 73.0\%$; $P < 0.001$) was observed for DMI (% of MBW), ECM yield (kg/d, % of BW, and % of MBW), N intake, milk N yield, and MNE. Publication bias ($P \leq 0.05$) was detected for ECM yield (kg/d and % of BW), N intake, and milk N yield, and a tendency ($P = 0.08$) for ECM yield (% of MBW). The funnel plots for selected estimated variables are presented in Figure 4E–F.

Meta-Regression and Subgroup Analyses of Observed Variables

Meta-regression analyses assessing the effect of the continuous covariates year of publication, DIM, and dietary concentrations (% of DM) of NDF, CP, and forage on the WMD of observed variables are presented in Table 6. Year of publication affected ($P = 0.009$) the WMD of DMI, showing that the difference between breeds decreased by 0.040 kg/d per each incremental increase in publication year in favor of Jerseys over Holsteins in the reduced model after the removal of covariates with $P > 0.10$ (here and in other relationships presented later). The covariate DIM widened ($P < 0.001$) the breed difference for BW by 0.38 kg per each incremental increase in DIM, when controlling for the dietary concentrations of NDF and CP in the model, favoring Holsteins over Jerseys. However, when DIM and CP content of the diet were held constant in the meta-regression, dietary concentration of NDF tended ($P = 0.06$) to reduce the WMD of BW by 1.58 kg with each percentage-unit increase in NDF, benefiting Jersey versus Holstein cows. Furthermore, when controlling for DIM and NDF level in the model, dietary CP concentration decreased ($P = 0.002$) the WMD of BW by 8.26 kg with each percentage-unit increase in CP content, also favoring Jerseys.

Days in milk decreased ($P = 0.002$) the breed difference for MY by 0.033 kg/d per each incremental increase in DIM, when the dietary concentration of NDF was held constant in the model, benefiting Jersey over Holstein cows (Table 6). The dietary concentration of NDF decreased ($P = 0.01$) the WMD of MY by 0.26 kg/d with each percentage-unit increase in NDF level, when the effect of DIM was controlled for in the model, also favoring Jerseys. Days in milk affected ($P < 0.001$) the WMD of milk fat yield, showing that the difference between breeds narrowed by 1.96 g/d per each incremental DIM in favor of Jerseys versus Holsteins as lactation progressed. Year of publication affected ($P = 0.005$) the WMD of milk true protein yield, with breed difference decreasing by 3.27 g/d per each incremental increase in publication year, benefiting Jersey over Holstein cows. The dietary concentration of NDF affected ($P < 0.001$) the WMD of milk lactose yield, with the difference between breeds widening by 33.7 g/d with each percentage-unit increase in NDF content, when forage level was held constant in the meta-regression, benefiting Holsteins at expense of Jerseys. However, when dietary NDF concentration was controlled for in the model, the breed difference for milk lactose yield narrowed ($P = 0.05$) by 1.52 g/d with each percentage-unit increase in forage level in the diet DM, favoring Jersey over Holstein cows. Year of publication increased ($P < 0.001$) the WMD of milk fat concentration by 0.16 g/kg per each incremental increase in publication year, when DIM was held constant in the meta-regression, favoring Jerseys over Holsteins. When controlling for year of publication in the model, DIM widened ($P < 0.001$) the WMD of milk fat concentration by 0.051 g/kg per each incremental increase in DIM in favor of Jer-

Table 4. Effect of breed on the observed variables DMI, BW, MY, milk composition, and NDFD

	Holstein		Jersey		Heterogeneity[3]		Funnel plot test[4]
Item	Mean (SD)	n[1]	WMD[2] (95% CI)	*P*-value	I^2 (%)	*P*-value	*P*-value
DMI, kg/d	21.0 (2.24)	75	−4.15 (−4.53, −3.75)	<0.001	96.8	<0.001	0.03
BW, kg	617 (57.6)	45	−169 (−180, −157)	<0.001	99.3	<0.001	0.50
MY, kg/d	31.8 (6.09)	72	−10.8 (−11.8, −9.77)	<0.001	98.7	<0.001	<0.01
Milk fat, g/d	1,161 (225)	44	−61.6 (−111, −12.5)	0.01	95.3	<0.001	<0.01
Milk true protein, g/d	1,038 (162)	44	−211 (−241, −182)	<0.001	97.0	<0.001	<0.001
Milk lactose, g/d	1,715 (103)	19	−584 (−629, −538)	<0.001	75.5	<0.001	0.85
Milk fat, g/kg	37.2 (5.69)	71	16.1 (14.6, 17.5)	<0.001	97.8	<0.001	0.07
Milk true protein, g/kg	32.5 (2.99)	70	6.98 (6.48, 7.48)	<0.001	97.3	<0.001	0.17
Milk lactose, g/kg	48.5 (1.44)	40	−0.91 (−1.27, −0.55)	<0.001	94.9	<0.001	0.36
NDFD, % of NDF intake	55.8 (11.1)	14	1.88 (0.66, 3.10)	<0.01	84.3	<0.001	0.08

[1]n = number of mean comparisons between Holstein (control) and Jersey cows.

[2]WMD = Jersey versus Holstein (control) cows.

[3]*P*-value for χ^2 (Q) test of heterogeneity; I^2 = proportion of total variation of size effect estimates due to heterogeneity.

[4]Egger's regression asymmetry test (Egger et al., 1997).

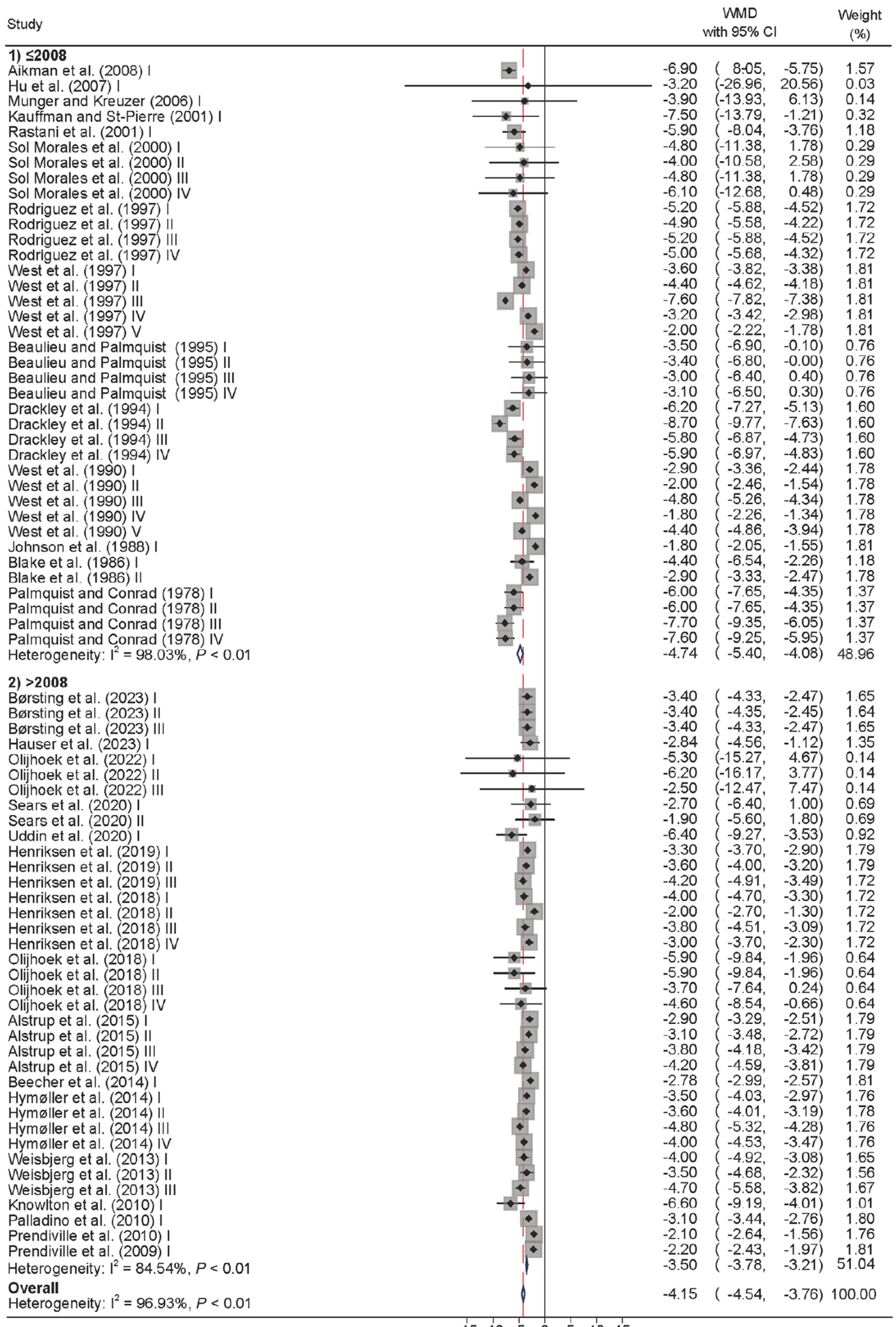

Figure 2. Forest plot with subgroup analyses for the effect of year of publication on the WMD of DMI (kg/d) between Holstein and Jersey cows. The horizontal axis shows the estimated WMD of each comparison represented by a black diamond within a gray box, which size is proportional to the relative weight of the comparison. Horizontal lines connected to each box represent the 95% CI. The empty diamond in the bottom represents the average WMD of each subgroup, and the red dashed vertical line is the overall WMD across comparisons. Roman numerals were added to each reference to distinguish multiple breed comparisons from the same study. Subgroups: (1) publication year ≤2008 and (2) publication year >2008. Note: weights and between-subgroup heterogeneity tests are from a random-effects model. References: Palmquist and Conrad (1978); Blake et al. (1986); Johnson et al. (1988); West et al. (1990, 1997); Drackley et al. (1994); Beaulieu and Palmquist (1995); Rodriguez et al. (1997); Sol Morales et al. (2000); Kauffman and St-Pierre (2001); Rastani et al. (2001); Munger and Kreuzer (2006); Hu et al. (2007); Aikman et al. (2008); Prendiville et al. (2009, 2010); Knowlton et al. (2010); Palladino et al. (2010); Weisbjerg et al. (2013); Beecher et al. (2014); Hymøller et al. (2014); Alstrup et al. (2015); Olijhoek et al. (2018, 2022); Henriksen et al. (2018, 2019); Sears et al. (2020); Uddin et al. (2020); Børsting et al. (2023); Hauser et al. (2023).

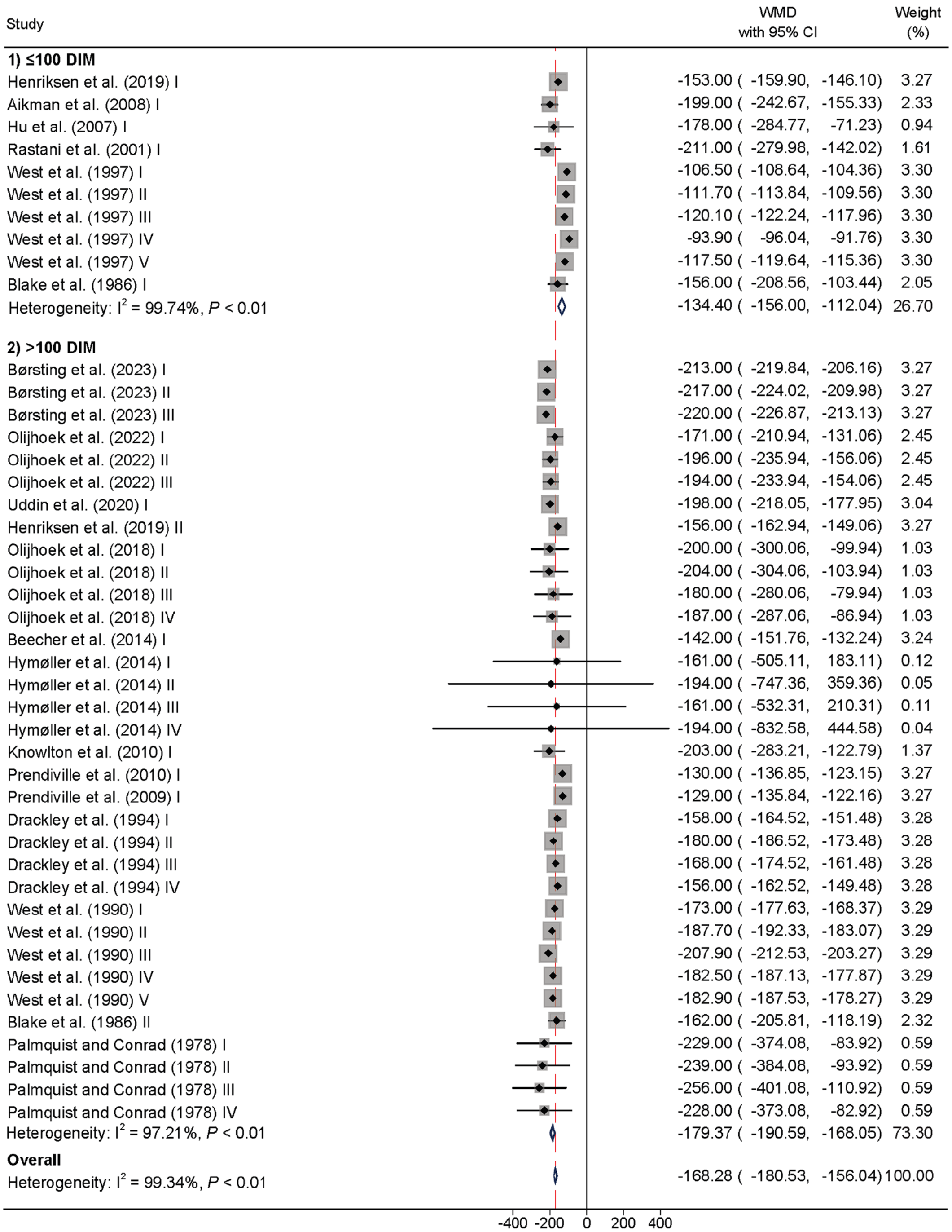

Figure 3. Forest plot with subgroup analysis for the effect of DIM on the WMD of BW (kg) between Holstein and Jersey cows. The horizontal axis shows the estimated WMD of each comparison represented by a black diamond within a gray box, which size is proportional to the relative weight of the comparison. Horizontal lines connected to each box represent the 95% CI. The empty diamond in the bottom represents the average WMD of each subgroup, and the red dashed vertical line is the overall WMD across comparisons. Roman numerals were added to each reference to distinguish multiple breed comparisons from the same study. Subgroups: (1) cows at ≤100 DIM and (2) cows at >100 DIM. Note: weights and between-subgroup heterogeneity tests are from a random-effects model. References: Palmquist and Conrad (1978); Blake et al. (1986); Drackley et al. (1994); West et al. (1990, 1997); Rastani et al. (2001); Hu et al. (2007); Aikman et al. (2008); Prendiville et al. (2009, 2010); Knowlton et al. (2010); Beecher et al. (2014); Hymøller et al. (2014); Olijhoek et al. (2018, 2022); Henriksen et al. (2019); Uddin et al. (2020); Børsting et al. (2023).

sey versus Holstein cows. Year of publication increased ($P < 0.001$) the WMD of milk true protein concentration by 0.11 g/kg per each incremental increase in publication year, when forage level was held constant in the meta-regression, benefiting Jerseys versus Holsteins. In contrast, the WMD of milk true protein concentration de-

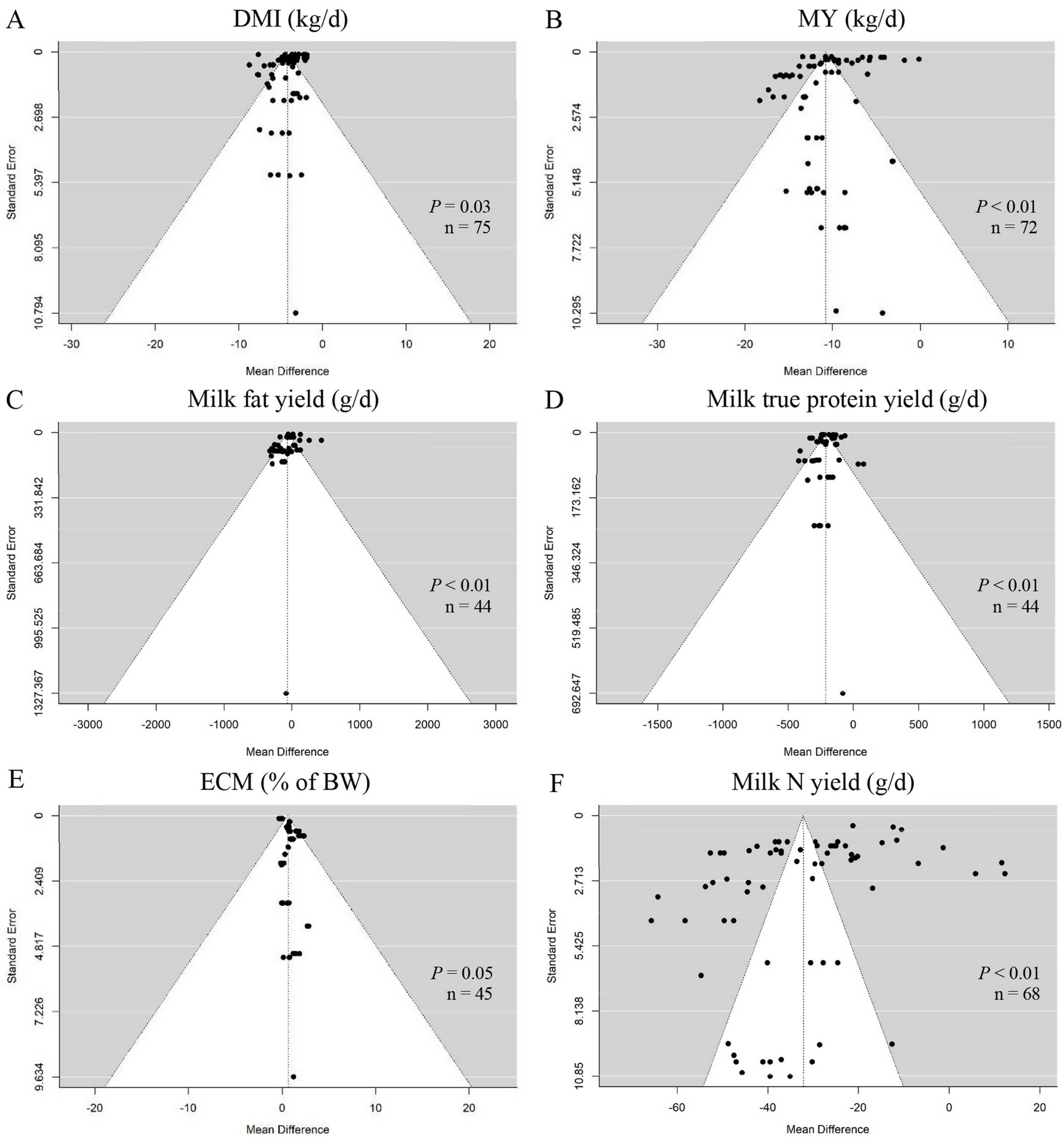

Figure 4. Selected funnel plots showing the effect of breed (Holsteins vs. Jerseys) on (A) DMI (kg/d), (B) MY (kg/d), (C) milk fat yield (g/d), (D) milk true protein yield (g/d), (E) ECM yield (% of BW), and (F) milk N yield (g/d). The horizontal axis indicates the WMD estimates, and the vertical axis indicates their corresponding SE. The *P*-value refers to the test of funnel plot asymmetry using the Egger's regression method (Egger et al., 1997); n = number of comparisons (i.e., observations) between Holstein and Jersey cows.

creased ($P = 0.002$) by 0.036 g/kg with each percentage-unit increase in dietary forage level, when controlling for year of publication in the model, favoring Holstein over Jersey cows. Year of publication widened ($P < 0.001$) the WMD of milk lactose concentration by 0.11 g/kg per each incremental increase in publication year, when CP

Table 5. Effect of breed on the estimated variables DMI, ECM yield, FE, N intake, milk N yield, and MNE

	Holstein		Jersey		Heterogeneity[3]		Funnel plot test[4]
Item	Mean (SD)	n[1]	WMD[2] (95% CI)	*P*-value	I^2 (%)	*P*-value	*P*-value
DMI, % of BW	3.42 (0.40)	50	0.44 (0.33, 0.54)	<0.001	33.1	0.26	0.41
DMI, % of MBW	17.1 (1.81)	50	0.64 (0.22, 1.05)	<0.01	94.8	<0.001	0.92
ECM yield, kg/d	30.9 (5.60)	68	−5.11 (−5.94, −4.29)	<0.001	98.0	<0.001	0.02
ECM yield, % of BW	5.02 (0.85)	45	0.64 (0.38, 0.90)	<0.001	73.0	<0.001	0.05
ECM yield, % of MBW	25.0 (3.76)	45	1.63 (0.60, 2.66)	<0.01	96.7	<0.001	0.08
FE_{MY}, kg/kg	1.50 (0.23)	72	−0.26 (−0.32, −0.20)	<0.001	0.00	1.00	0.42
FE_{ECM}, kg/kg	1.45 (0.20)	68	0.02 (−0.04, 0.08)	0.56	0.00	1.00	0.31
N intake, g/d	582 (80.1)	74	−115 (−127, −103)	<0.001	100	<0.001	<0.01
Milk N yield, g/d	163 (31.1)	68	−32.2 (−36.2, −28.1)	<0.001	99.1	<0.001	<0.01
MNE, % of N intake	27.9 (5.60)	68	−0.30 (−0.91, 0.31)	0.33	98.8	<0.001	0.41

[1]n = number of mean comparisons between Holstein (control) and Jerseys cows.

[2]WMD = Jersey versus Holstein (control) cows.

[3]*P*-value for χ^2 (Q) test of heterogeneity; I^2 = proportion of total variation of size effect estimates due to heterogeneity.

[4]Egger's regression asymmetry test (Egger et al., 1997).

level was kept constant in the meta-regression, benefiting Holsteins versus Jerseys. In contrast, the WMD of milk lactose concentration narrowed (P = 0.02) by 0.15 g/kg with each percentage-unit increase in dietary CP level, when the effect of year of publication was controlled for in the model, benefiting Jerseys at expense of Holsteins.

Subgroup analyses for observed variables are shown in Figures 5 and 6. The WMD of DMI was lower in Jerseys versus Holsteins but the difference between breeds narrowed in newer (>2008; −3.49 k/d) versus older (≤2008; −4.73 kg/d) publications (test of moderator: P = 0.003), favoring Jersey cows (Figure 5A). Similarly, the WMD of milk true protein yield was lower in Jersey versus Holstein cows, with the gap between breeds decreasing in more recent (>2008; −166 g/d) versus older (≤−239 g/d) publications (test of moderator: P = 0.01), benefiting Jerseys (Figure 5B). In addition, the WMD of milk fat (Figure 5C; test of moderator: P < 0.001) and milk true protein concentrations (Figure 5D; test of moderator: P < 0.001) also favored Jerseys over Holsteins in recent publications: +13.2 and +19.5 g/kg (milk fat) and +5.99 and +8.15 g/kg (milk true protein), in studies published ≤2008 and >2008, respectively. The WMD of milk lactose concentration benefited Jerseys (+0.73 g/kg) in studies published ≤2008 but favored Holsteins (−1.15 g/kg) in recent studies (i.e., >2008; Figure 5E; test of moderator: P < 0.001). Our results further showed that the WMD of BW favored Holstein over Jersey cows with the progress in DIM: −134 and −179 kg at ≤100 DIM and >100 DIM, respectively (Figure 6A; test of moderator: P < 0.001). Furthermore, the WMD of milk fat yield tended to favor Jersey over Holstein cows as lactation progressed: −103 and −49.4 g/d at ≤100 DIM and >100 DIM, respectively (Figure 6B; test of moderator: P = 0.09). In line with the WMD of milk fat yield, the WMD of milk fat concentration favored Jersey over Holstein cows as lactation progressed: +12.0 and +18.0 g/kg at ≤100 DIM and >100 DIM, respectively (Figure 6C; test of moderator: P < 0.001).

Meta-regression analyses examining the effect of forage type on the WMD of observed variables are presented in Table 7. Figure 7 also shows the individual effect of forage type on each WMD in the subset of comparisons in which each forage type was fed to cows. Note that the effect on milk lactose concentration is not shown in Table 7 and Figure 7 because of the unbalanced number of comparisons across the forage types (n = 5, n = 4, and n = 31 for corn silage, alfalfa/grass-clover mix, and others, respectively). The reader should be aware that nonsignificant coefficients for the forage types alfalfa/grass-clover mix and others reported in Table 7 are for the comparisons against corn silage (i.e., the intercept of the meta-regression) and this does not mean that the WMD of a given variable is not significant (e.g., see coefficient of BW for cows fed alfalfa/grass-clover mix in Table 7 and Figure 7).

The intercept corresponding to corn silage–based diets revealed that the difference between breeds was not significant for the WMD of milk fat yield (Table 7). However, the difference between breeds favored (P < 0.001) Holsteins over Jerseys for DMI (WMD = −3.49 kg/d), BW (WMD = −160 kg), MY (WMD = −8.77 kg/d), and milk true protein yield (WMD = −195 g/d). In contrast, the difference between breeds favored (P < 0.001) Jerseys versus Holsteins for the concentrations of milk fat (WMD = +13.1 g/kg) and milk true protein (WMD = +5.90 g/kg).

In comparison with cows fed corn silage diets, feeding cows alfalfa/grass-clover mix diets affected (P ≤ 0.05) the difference between breeds for DMI (WMD = −2.46 kg/d), MY (WMD = −5.13 kg/d), and yields of milk fat (WMD = −137 g/d) and milk true protein (WMD = −107

Table 6. Meta-regression of the effect of the continuous covariates year of publication, DIM, and dietary concentrations of NDF, CP, and forage on the WMD of the observed variables DMI, BW, MY, and milk composition in Holstein versus Jersey cows

Dependent variable (WMD)	Meta-regression parameter[1]							
	Intercept	Year[2]	DIM[3]	NDF,[4] % of DM	CP,[4] % of DM	Forage, % of DM	Adjusted R^2 (%)[5]	n[6]
DMI, kg/d	−81.0 (29.6)**	0.040 (0.014)**	NS	NS	NS	NS	8.36	75
BW, kg	−315 (41.1)***	NS	−0.38 (0.075)***	1.58 (0.85)†	8.26 (2.66)**	NS	67.0	44
MY, kg/d	−24.0 (4.02)***	NS	0.033 (0.011)**	0.26 (0.11)*	NS	NS	19.1	71
Milk fat, g/d	−315 (64.9)***	NS	1.96 (0.49)***	NS	NS	NS	35.6	43
Milk true protein, g/d	−6,760 (2,332)**	3.27 (1.16)**	NS	NS	NS	NS	11.7	44
Milk lactose, g/d	399 (159)*	NS	NS	−33.7 (5.50)***	NS	1.52 (0.78)*	100	18
Milk fat, g/kg	−320 (89.8)***	0.16 (0.045)***	0.051 (0.012)***	NS	NS	NS	45.2	70
Milk true protein, g/kg	−203 (28.9)***	0.11 (0.015)***	NS	NS	NS	−0.036 (0.011)**	54.2	69
Milk lactose, g/kg	215 (35.8)***	−0.11 (0.018)***	NS	NS	0.15 (0.066)*	NS	59.5	40

[1]Covariate effects: †$P \leq 0.10$, *$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$, NS = nonsignificant; values within parentheses represent the SE for a given meta-regression coefficient.

[2]Year of publication ranged from 1978 to 2023.

[3]Average DIM was calculated by averaging the DIM on the first and last day of each study.

[4]The dietary concentrations of NDF and CP were back-calculated from NDF and CP intake when not reported in the selected publications.

[5]Adjusted R^2 = proportion of the between-study variance (heterogeneity) explained by the continuous covariates.

[6]n = number of mean comparisons (i.e., observations) between Holstein and Jersey cows.

g/d) as shown in Table 7. When the meta-regressions were conducted in the subset of comparisons in which alfalfa/grass-clover mix was fed to cows (Figure 7), the WMD of DMI, MY, and milk true protein yield averaged, respectively, −5.96 kg/d (n = 14), −14.0 kg/d (n = 14), and −299 g/d (n = 9; data not shown), further widening ($P < 0.001$) the breed differences for all these variables, favoring Holstein over Jersey cows. Compared with cows fed corn silage diets, feeding the forage type others affected ($P < 0.001$) the difference between breeds for the concentrations of milk fat (WMD = +5.16 g/kg) and milk true protein (WMD = +1.96 g/kg). In addition, the forage type others tended ($P = 0.06$) to affect the breed differences for MY (WMD = −2.04 kg/d). When the meta-regressions were conducted in the subset of comparisons in which the forage type others was fed to cows (Figure 7), the WMD of MY, milk fat concentration, and milk true protein concentration averaged, respectively, −10.8 kg/d (n = 35), +18.3 g/kg (n = 34), and +7.87 g/kg (n = 34; data not shown), further widening ($P < 0.001$) the breed difference for all 3 variables, favoring Holsteins for MY but Jerseys for milk fat and milk true protein contents.

Meta-Regression and Subgroup Analyses of Estimated Variables

Meta-regression analyses evaluating the effect of the continuous covariates year of publication, DIM, and dietary concentrations (% of DM) of NDF, CP, and forage on the WMD of estimated variables are presented in Table 8. Days in milk affected ($P \leq 0.03$) the WMD of DMI (+0.0021% of BW and +0.015% of MBW), ECM yield (+0.046 kg/d), and milk N yield (+0.10 g/d), with the difference between breeds either widening (DMI [% of BW and % of MBW]) or narrowing (ECM yield and milk N yield) with incremental increase in DIM, favoring Jersey over Holstein cows. In addition, DIM increased ($P < 0.001$) the breed difference for ECM yield (% of BW) by 0.012% per each incremental increase in DIM, when the dietary NDF concentration was held constant in the model, also benefiting Jerseys versus Holsteins. However, with DIM controlled for in the meta-regression, dietary level of NDF narrowed ($P = 0.002$) the breed difference for ECM yield (% of BW) by 0.026% with each percentage-unit increase in the concentration of NDF in the diet DM, benefiting Holsteins over Jerseys. The covariate year of publication widened ($P < 0.001$) the difference between breeds for ECM yield (% of MBW) by 0.083% per each incremental increase in publication year, when DIM and forage level were held constant in the model, favoring Jersey at expense of Holstein cows. When year of publication and dietary forage level were kept constant in the meta-regression, DIM increased ($P < 0.001$) the WMD of ECM yield (% of MBW) by 0.056%

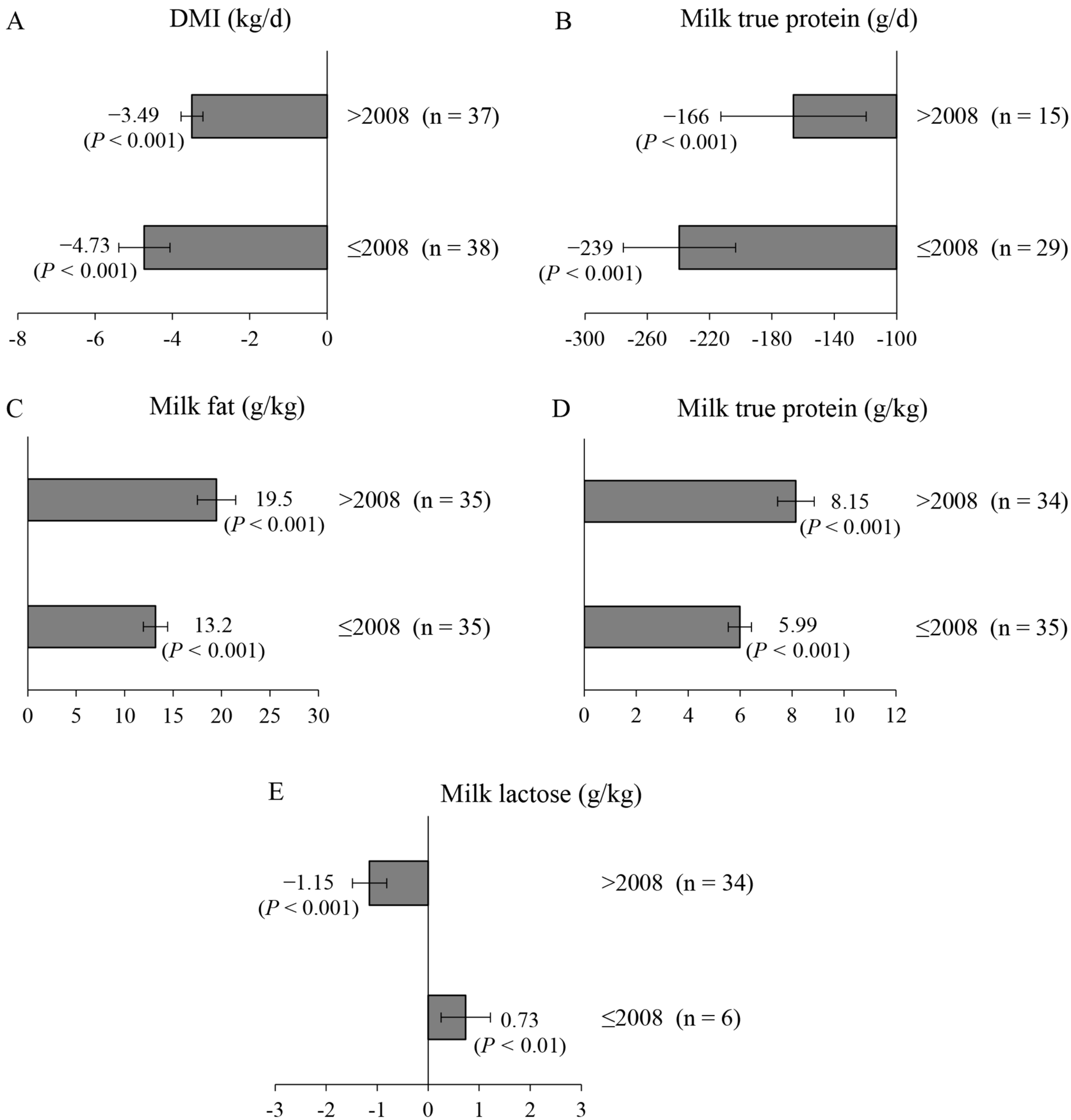

Figure 5. Subgroup analyses on the effect of year of publication on the WMD (95% CI) between Holstein and Jersey cows for the following observed variables: (A) DMI (kg/d), test of moderator to detect group differences ($P = 0.003$); (B) milk true protein yield (g/d), test of moderator to detect group differences ($P = 0.01$); (C) milk fat concentration (g/kg), test of moderator to detect group differences ($P < 0.001$); (D) milk true protein concentration (g/kg), test of moderator to detect group differences ($P < 0.001$); and (E) milk lactose concentration (g/kg), test of moderator to detect group differences ($P < 0.001$). Subgroups: (1) publication year ≤2008 and (2) publication year >2008. The horizontal error bars show the SD for each subgroup.

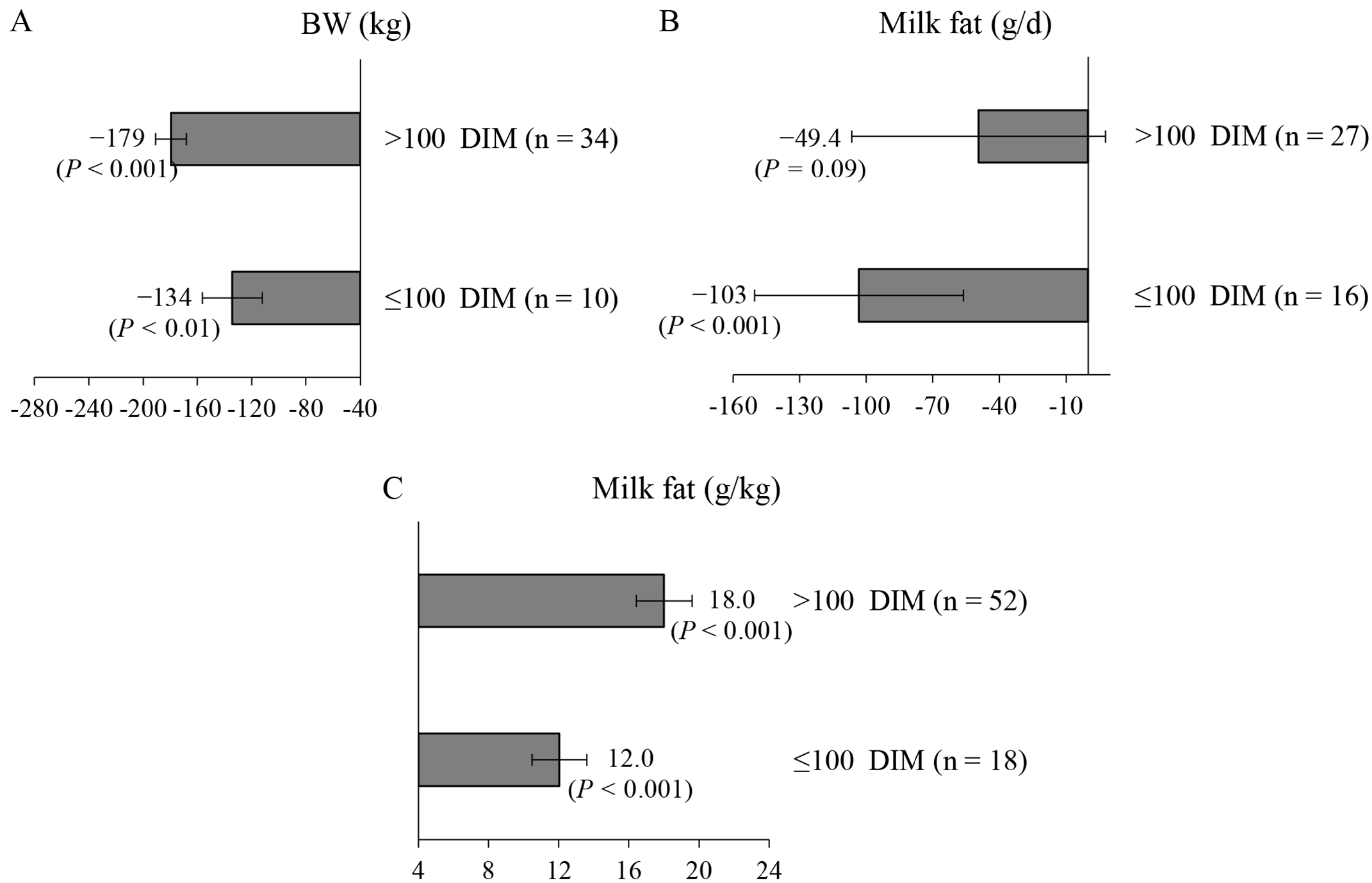

Figure 6. Subgroup analyses on the effect of DIM on the WMD (95% CI) between Holstein and Jerseys cows for the following observed variables: (A) BW (kg), test of moderator to detect group differences ($P < 0.001$); (B) milk fat yield (g/d), test of group differences to detect group differences ($P = 0.09$); and (C) milk fat concentration (g/kg), test of moderator to detect group differences ($P < 0.001$). Subgroups: (1) cows at ≤100 DIM, and (2) cows at >100 DIM. The horizontal error bars show the SD for each subgroup.

Table 7. Meta-regression of the effect of the categorical covariate forage type on the WMD of the observed variables DMI, BW, MY, and milk composition in Holstein versus Jersey cows[1]

	Meta-regression parameter[2]				
Dependent variable (WMD)	Corn silage	Alfalfa/grass-clover mix	Others	Adjusted R^2 (%)[3]	n[4]
DMI, kg/d	−3.49 (0.31)***	−2.46 (0.53)***	−0.44 (0.39)	25.1	75
BW, kg	−160 (9.44)***	−13.0 (15.8)	−15.9 (14.0)	0.00	45
MY, kg/d	−8.77 (0.86)***	−5.13 (1.41)***	−2.04 (1.07)†	19.0	72
Milk fat, g/d	−64.8 (45.2)	−137 (69.3)*	58.1 (54.7)	19.8	44
Milk true protein, g/d	−195 (29.7)***	−107 (40.1)**	15.1 (34.2)	31.0	44
Milk fat, g/kg	13.1 (1.20)***	1.76 (2.16)	5.16 (1.54)***	13.1	71
Milk true protein, g/kg	5.90 (0.44)***	−0.41 (0.77)	1.96 (0.56)***	21.9	71

[1]Forage type: (1) corn silage–based diets (“corn silage”), (2) alfalfa hay– or alfalfa silage–based diets and grass-clover mix silage (“alfalfa/grass-clover mix”), and (3) forage sources that did not fit in any of the other 2 forage types (“others”). Forage types 1 and 2 were defined based on the forage type that accounted for at least 55% of the forage component in the diet DM; forage type 3 included diets with ≥2 forage sources at equal or similar inclusion levels in the diet DM, but below the minimum 55% forage DM threshold, as well as forage sources with low number of observations (i.e., freshly-cut or grazed herbage; n = 4).

[2]Covariate effects: †$P \le 0.10$, *$P \le 0.05$, **$P \le 0.01$, ***$P \le 0.001$; values within parentheses represent the SE for a given meta-regression coefficient. Note: The meta-regression coefficients of the forage type corn silage represent the true coefficients (i.e., intercepts), and the meta-regression coefficients of the forage types alfalfa/grass-clover mix and others represent the difference between either of these 2 forage types and corn silage.

[3]Adjusted R^2 = proportion of the between-study variance (heterogeneity) explained by the categorical covariates.

[4]n = number of mean comparisons (i.e., observations) between Holstein and Jersey cows.

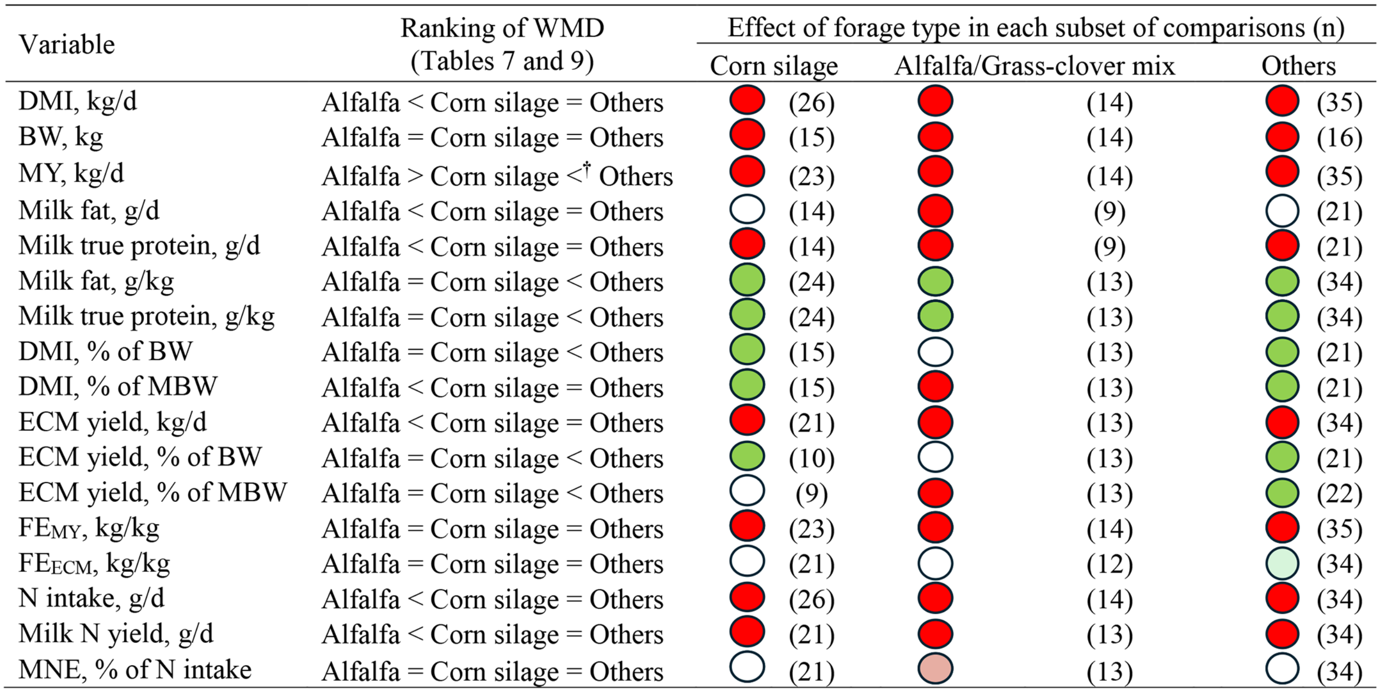

Figure 7. Ranking of the effect of forage type on the WMD of different observed and estimated variables in comparisons reported in Tables 7 and 9. Forage type: (1) corn silage–based diets ("corn silage"), (2) alfalfa hay– or alfalfa silage–based diets and grass-clover mix silage ("alfalfa/grass-clover mix"), and (3) forage sources that did not fit in any of the other 2 forage types ("others"). Forage types 1 and 2 were defined based on the forage type that accounted for at least 55% of the forage component in the diet DM; forage type 3 includes diets with ≥2 forage sources at equal or similar inclusion levels in the diet DM, but below the minimum 55% forage DM threshold, as well as forage sources with low number of observations (i.e., freshly-cut or grazed herbage; n = 4). Within individual row (visualized by the circles), the effect of each forage type was tested in the subset of comparisons in which it was fed, with the number of comparisons shown between parentheses. The open circles are for nonsignificant WMD. The green and red circles are, respectively, for positive (favoring Jerseys over Holsteins) or negative (favoring Holsteins over Jerseys) WMD with $P \leq 0.05$, whereas pale green (FE_{ECM}) and pale red (MNE) circles correspond to tendencies. †$P \leq 0.10$.

per each incremental increase in DIM, also benefiting Jerseys over Holsteins. In contrast, when controlling for year of publication and DIM in the model, dietary forage level narrowed ($P = 0.002$) the gap between breeds by 0.040% with each percentage-unit increase in the concentration of forage in the diet DM, favoring Holstein versus Jersey cows.

The dietary concentration of NDF narrowed ($P = 0.02$) the gap between breeds for FE_{MY} by 0.014 kg/kg with each percentage-unit increase in NDF level in the diet DM, favoring Jerseys over Holsteins (Table 8). Year of publication narrowed ($P = 0.005$) the gap between breeds for N intake by 1.16 g/d per each per each incremental increase in publication year, when dietary CP concentration was held constant in the meta-regression, benefiting Jersey versus Holstein cows. In contrast, dietary CP concentration further increased ($P = 0.003$) the breed difference in N intake by 9.48 g/d with each percentage-unit increase in CP level in the diet DM, when year of publication was controlled for in the model, favoring Holsteins at the expense of Jerseys (Table 8). Our meta-regression also showed that none of the covariates evaluated herein affected FE_{ECM} and MNE (Table 8). Therefore, their intercept in the meta-regression model corresponds to the WMD reported in Table 5.

Subgroup analyses conducted for estimated variables are shown in Figures 8 and 9. The WMD of ECM yield (% of MBW) favored Jersey over Holstein cows in recent publications: −0.13 and +3.30% of MBW in studies published ≤2008 and >2008, respectively (Figure 8A; test of moderator: $P < 0.001$). Figure 8B further revealed that N intake was lower in Jerseys versus Holsteins but the gap between breeds narrowed in newer (>2008; −99.1 g/d) versus older (≤2008; −132 g/d) publications (test of moderator: $P = 0.004$), favoring Jersey cows. The WMD of DMI (% of BW) and DMI (% of MBW) favored Jersey versus Holstein cows as the lactation progressed: +0.26 and +0.49% of BW (Figure 9A; test of moderator: $P = 0.03$), and −0.64 and +0.83% of MBW (Figure 9B; test of moderator: $P = 0.009$) at ≤100 DIM and >100 DIM, respectively. Similarly, ECM yield (kg/d) was lower in Jerseys versus Holsteins, with the breed difference shrinking as the lactation advanced, benefiting Jersey cows (Figure 9C): −6.61 and −4.63 kg/d at ≤100 DIM and >100 DIM, respectively (test of moderator: $P = 0.03$). The subgroup analysis for the WMD of ECM yield (% of BW and %

Table 8. Meta-regression of the effect of the continuous covariates of year of publication, DIM, dietary concentrations of NDF, CP, and forage on the WMD of the estimated variables DMI, ECM yield, FE, N intake, milk N yield, and MNE in Holstein versus Jersey cows

Dependent variable (WMD)	Meta-regression parameter[1]							
	Intercept	Year[2]	DIM[3]	NDF,[4] % of DM	CP,[4] % of DM	Forage, % of DM	Adjusted R^2 (%)[5]	n[6]
DMI, % of BW	0.14 (0.13)	NS	0.0021 (0.0009)*	NS	NS	NS	16.5	50
DMI, % of MBW	−1.69 (0.64)**	NS	0.015 (0.0043)***	NS	NS	NS	25.6	51
ECM yield, kg/d	−11.4 (1.13)***	NS	0.046 (0.0078)***	NS	NS	NS	43.2	67
ECM yield, % of BW	−0.062 (0.328)	NS	0.012 (0.0012)***	−0.026 (0.0084)**	NS	NS	100	45
ECM yield, % of MBW	−171 (42.7)***	0.083 (0.021)***	0.056 (0.0059)***	NS	NS	−0.040 (0.013)**	84.0	45
FE_{MY}, kg/kg	−0.75 (0.22)***	NS	NS	0.014 (0.0060)*	NS	NS	0.00	72
FE_{ECM}, kg/kg	0.012 (0.036)	NS	NS	NS	NS	NS	0.00	65
N intake, g/d	−2,288 (835)**	1.16 (0.41)**	NS	NS	−9.48 (3.21)**	NS	17.1	73
Milk N yield, g/d	−46.3 (6.45)***	NS	0.10 (0.045)*	NS	NS	NS	8.23	68
MNE, % of N intake	−0.30 (0.31)	NS	NS	NS	NS	NS	2.10	68

[1]Covariate effects: *$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$, NS = nonsignificant; values within parentheses represent the SE for a given meta-regression coefficient.
[2]Year of publication ranged from 1978 to 2023.
[3]Average DIM was calculated by averaging the DIM on the first and last day of each study.
[4]The dietary concentrations of NDF and ADF were back-calculated from NDF and CP intake when not reported in the selected publications.
[5]Adjusted R^2 = proportion of the between-study variance (heterogeneity) explained by the continuous covariates.
[6]n = number of mean comparisons (i.e., observations) between Holstein and Jersey cows.

of MBW) also favored Jersey compared with Holstein cows as lactation progressed: −0.070 and +0.92% of BW (Figure 9D; test of moderator: $P < 0.001$) and −1.94 and +2.54% of MBW (Figure 9E; test of moderator: $P < 0.001$) at ≤100 DIM and >100 DIM, respectively.

The effects of the categorical variable forage type on the WMD of estimated variables are presented in Table 9 and Figure 7. The intercept corresponding to corn silage–based diets revealed that the difference between breeds was not significant for the WMD of ECM yield (% of BW and % of MBW), FE_{ECM}, and MNE. In corn silage–based diets, the difference between breeds favored Jerseys over Holsteins for DMI (% of BW [WMD = +0.27%; $P < 0.001$] and % of MBW [WMD = +0.74%; $P = 0.01$]) and Holsteins over Jerseys ($P < 0.001$) for ECM yield (WMD = −4.18 kg/d), FE_{MY} (WMD = −0.28 kg/kg), N intake (WMD = −102 g/d), and milk N yield (WMD = −27.3 g/d).

In comparison with cows fed corn silage diets, feeding cows with alfalfa/grass-clover mix affected ($P < 0.001$) the difference between breeds for DMI (% of MBW), ECM yield (kg/d), N intake, and milk N yield (Table 9). When the meta-regressions were conducted in the subset of comparisons in which alfalfa/grass-clover mix was fed to cows (Figure 7), the WMD of ECM yield (kg/d), N intake, and milk N yield averaged, respectively, −8.20 kg/d (n = 13), −172 g/d (n = 14), and −47.1 g/d (n = 13; data not shown), further widening ($P < 0.001$) the breed difference for these 3 variables, benefiting Holstein over Jersey cows. Furthermore, the WMD of DMI (% of MBW) favored Jersey versus Holstein cows when feeding corn silage (+0.74% of MBW; $P = 0.01$) but the opposite was observed in cows fed alfalfa/grass-clover mix (−1.99% of MBW; $P = 0.001$; Table 9 and Figure 7). Compared with cows fed corn silage, feeding the forage type others affected ($P \leq 0.03$) the difference between breeds for DMI (% of BW) and ECM yield (% of BW and % of MBW) as shown in Table 9. When the meta-regressions were conducted in the subset of comparisons in which the forage type others was fed to cows (Figure 7), the WMD of DMI (% of BW), ECM yield (% of BW), and ECM yield (% of MBW) averaged, respectively, +0.48% (n = 21), +0.89% (n = 21), and +2.90% (n = 22; data not shown), further widening ($P < 0.001$) the breed difference for these 3 variables, favoring Jerseys over Holsteins.

DISCUSSION

The current meta-analysis aimed to compare DMI, production performance, FE, and MNE in Holstein versus Jerseys cows by integrating data from studies in which both breeds were evaluated under similar experimental conditions. This approach was used to minimize potential

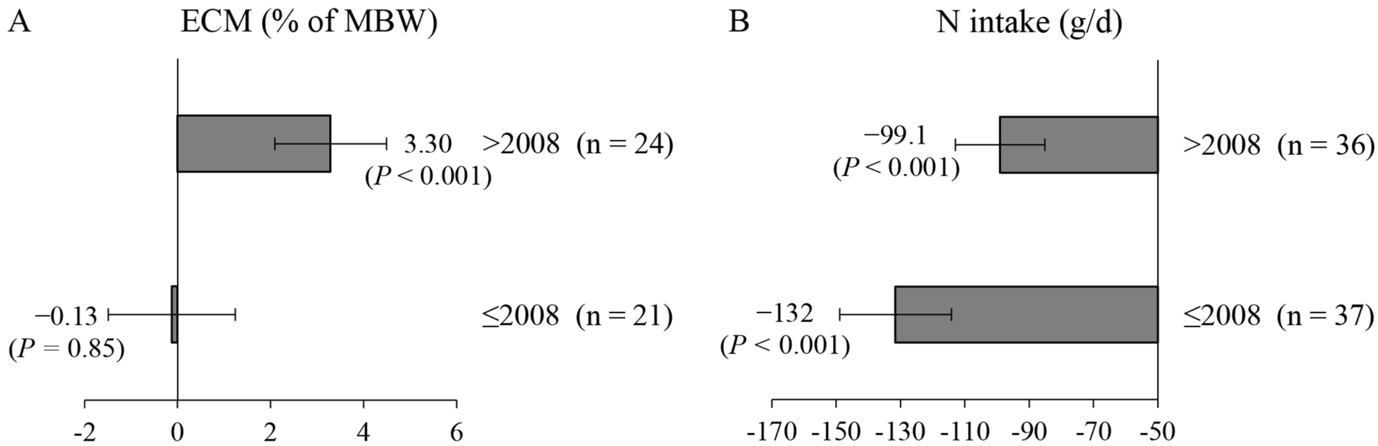

Figure 8. Subgroup analyses on the effect of year of publication on the WMD (95% CI) between Holstein and Jersey cows for the following estimated variables: (A) ECM yield (% of MBW), test of moderator to detect group differences ($P < 0.001$); and (B) N intake (g/d), test of moderator to detect group differences ($P = 0.004$). Subgroups: (1) publication year ≤2008 and (2) publication year >2008. The horizontal error bars show the SD for each subgroup.

biases arising from differences in experimental design, data collection, and study location and duration, thereby ensuring accurate comparisons between these 2 breeds.

Production Performance Variables

Jerseys consumed 4.15 kg/d less DM than Holsteins. However, when DMI was expressed as a proportion of BW and MBW, it was greater in Jersey versus Holstein cows. Compared with Holsteins, Jerseys have a heavier reticulorumen and total gastrointestinal tract per unit of BW (Beecher et al., 2014), showing greater relative capacity to consume feed (Prendiville et al., 2009, 2010; Sears et al., 2020). In addition, Aikman et al. (2008) reported that digesta passage through the digestive tract was faster in Jerseys than Holsteins, possibly because the total chewing time (i.e., eating plus rumination) per unit of NDF was, on average, 43% longer in Jersey than Holstein cows, resulting in smaller feed particles and less retention time in the rumen of Jerseys. Both MY and ECM yield (kg/d) were lower in Jersey than Holstein cows, which can be explained by reduced DMI in the Jersey breed. However, ECM yield (% of BW and % of MBW) was greater in Jerseys compared with Holsteins, in line with increased DMI (% of BW and % of MBW) in Jersey cows. Note that the average MY of Holsteins (31.8 kg/d) and Jerseys (21.1 kg/d) included in the present meta-analysis was 14.1% and 21.9% lower, respectively, than current MY for these 2 breeds, which averaged 37.0 kg/d for Holsteins and 27.0 kg/d for Jerseys, as shown by Olthof et al. (2023) using data obtained from 3 commercial dairy farms in Michigan. Therefore, our results should be interpreted within the constraints of the datasets used in the present meta-analysis because changes in management (e.g., animal health, animal welfare, nutrition, breeding/genetics) may not be symmetrical between Holsteins and Jerseys over time.

Yields of milk fat, milk true protein, and milk lactose were lower in Jerseys compared with Holsteins, in agreement with the lowest DMI (kg/d) and MY in the Jersey breed. One could hypothesize that the differences in milk component yields in Jerseys versus Holstein cows are linked to breed-specific metabolism of nutrients in the mammary gland. However, we are not aware of any research that has investigated this hypothesis either mechanistically or through nutrigenomics. Regardless, Alstrup et al. (2015) observed no effect of breed (Holsteins vs. Jerseys) on the jugular-mammary venous difference and efficiency of mammary extractions of glucose, NEFA, BHB, lactate, triglycerides, urea, and cholesterol in diets with rumen-protected vegetable fat and a mix of rumen-protected vegetable fat and hydroxy-Met-analog-isobutyrate during early, mid, and late lactation. In an earlier study, Hu et al. (2007) showed that whereas the plasma concentrations of creatinine (−15.5%) and glucose (−7.64%) were lower in Jersey than Holstein cows, NEFA levels in plasma increased by 87.2% in Jerseys with feeding diets containing different levels of DCAD during early lactation. In contrast, Rastani et al. (2001) observed that the plasma concentration of NEFA was lower in Jerseys (mean = 277 μEq/L) than Holsteins (mean = 503 μEq/L) at wk 1 of lactation, in line with data from French (2006), who reported mean plasma NEFA values of 521 and 775 μEq/L from 3 d prepartum to 1 d postpartum in Jersey and Holstein cows, respectively. Note that Rastani et al. (2001) did not detect breed differences in the plasma concentration of NEFA from wk 3 to wk 17 of lactation. Hu et al. (2007) also observed that

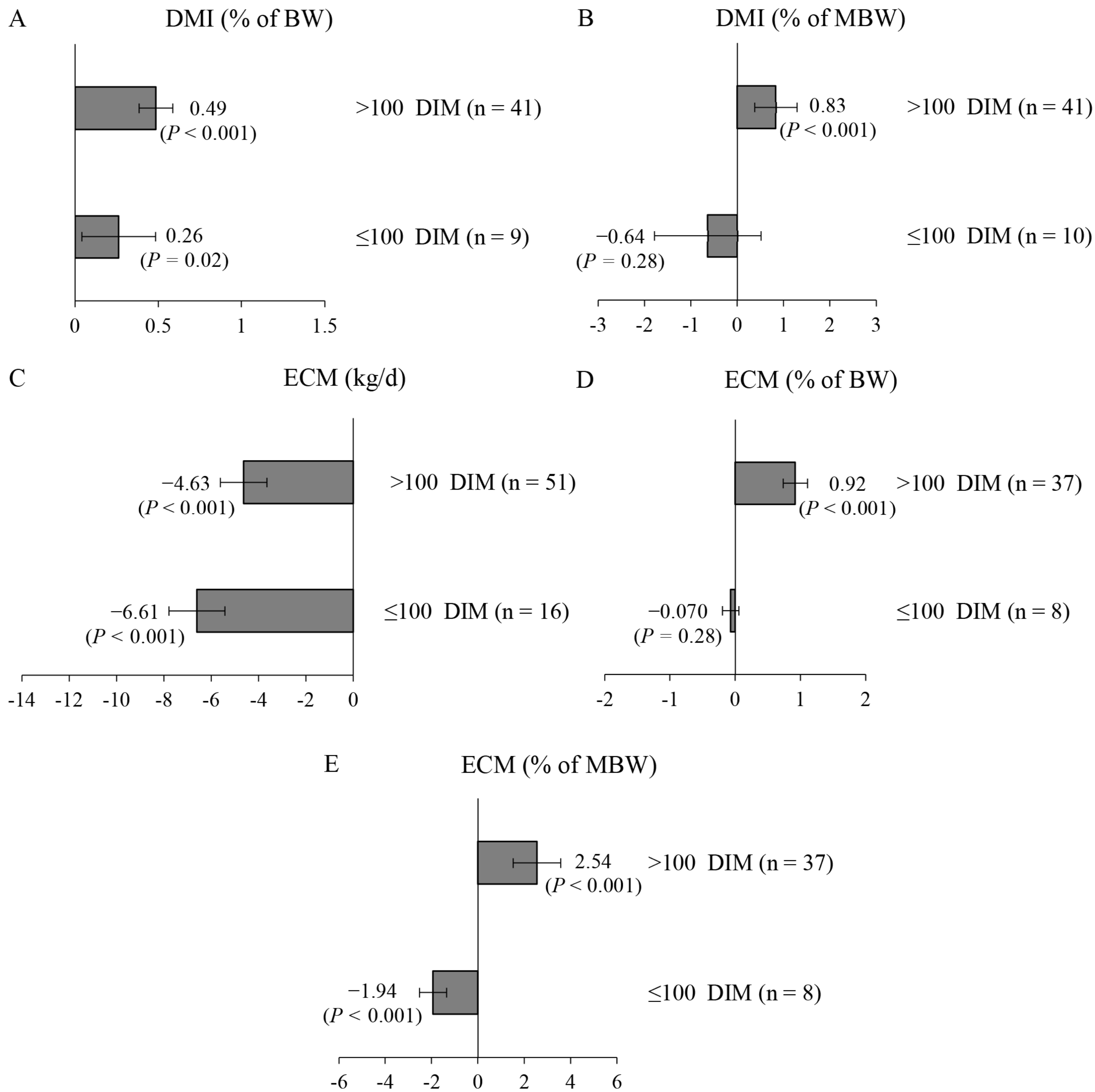

Figure 9. Subgroup analyses on the effect of DIM on the WMD (95% CI) between Holstein and Jersey cows for the following estimated variables: (A) DMI (% of BW), test of moderator to detect group differences ($P = 0.03$); (B) DMI (% of MBW), test of moderator to detect group differences ($P = 0.009$); (C) ECM yield (kg/d), test of moderator to detect group differences ($P = 0.03$); (D) ECM yield (% of BW), test of moderator to detect group differences ($P < 0.001$); and (E) ECM yield (% of MBW), test of moderator to detect group differences ($P < 0.001$). Subgroups: (1) cows at ≤100 DIM and (2) cows at >100 DIM. The horizontal error bars show the SD for each subgroup.

the plasma concentration of Arg decreased (−9.64%) and that of His increased (+18.8%) in Jersey versus Holstein cows despite no other breed effects in the concentration of individual EAA in plasma. Overall, data from Rastani et al. (2001), French (2006), and Hu et al. (2007) support breed differences in nutrient metabolism and use of body reserves during the negative energy balance in early lactation. Therefore, the possibility that breed-

Table 9. Meta-regression of the effect of the categorical covariate forage type on the WMD of the estimated variables DMI, ECM yield, FE, N intake, milk N yield, and MNE in Holstein versus Jersey cows[1]

	Meta-regression categorial covariate[2]				
Dependent variable (WMD)	Corn silage	Alfalfa/grass-clover mix	Others	Adjusted R^2 (%)[3]	n[4]
DMI, % of BW	0.27 (0.062)***	−0.25 (0.34)	0.21 (0.085)*	97.1	49
DMI, % of MBW	0.74 (0.30)*	−1.99 (0.55)***	0.44 (0.40)	32.8	49
ECM yield, kg/d	−4.18 (0.74)***	−4.16 (1.21)***	−0.42 (0.90)	20.6	68
ECM yield, % of BW	0.026 (0.062)	0.050 (0.13)	0.87 (0.12)***	100	44
ECM yield, % of MBW	0.45 (0.94)	−1.60 (1.40)	2.38 (1.12)*	20.7	44
FE_{MY}, kg/kg	−0.28 (0.049)***	−0.056 (0.11)	0.057 (0.065)	0.00	72
FE_{ECM}, kg/kg	−0.020 (0.050)	0.070 (0.27)	0.10 (0.065)	0.00	67
N intake, g/d	−102 (8.45)***	−70.7 (14.3)***	0.026 (11.3)	28.2	74
Milk N yield, g/d	−27.3 (3.48)***	−19.6 (5.63)***	−2.33 (4.36)	16.5	68
MNE, % of N intake	0.037 (0.61)	−0.85 (0.95)	−0.32 (0.74)	0.00	68

[1]Forage type: (1) corn silage–based diets ("corn silage"), (2) alfalfa hay– or alfalfa silage–based diets and grass-clover mix silage ("alfalfa/grass-clover mix"), and (3) forage sources that did not fit in any of the other 2 forage types ("others"). Forage types 1 and 2 were defined based on the forage type that accounted for at least 55% of the forage component of the diet DM; forage type 3 includes diets with ≥2 forage sources at equal or similar inclusion levels in the diet DM, but below the minimum 55% forage DM threshold, as well as forage sources with low number of observations (i.e., freshly-cut or grazed herbage; n = 4). Note: The meta-regression coefficients of the forage type corn silage represent the true coefficients (i.e., intercepts), and the meta-regression coefficients of the forage types alfalfa/grass-clover mix and others represent the difference between either of these 2 forage types and corn silage.

[2]Covariate effects: *$P \leq 0.05$, ***$P \leq 0.001$; values within parentheses represent the SE for a given meta-regression coefficient.

[3]Adjusted R^2 = proportion of the between-study variance (heterogeneity) explained by the categorical covariates.

[4]n = number of mean comparisons (i.e., observations) between Holstein and Jersey cows.

specific postabsorptive metabolism of nutrients could be involved in the milk component output differences seen between Jerseys and Holsteins cannot be excluded and warrants further investigations.

Milk fat and milk true protein concentrations were greater in Jerseys than Holsteins possibly because these 2 milk components became more concentrated because of the lower milk volume in Jersey versus Holstein cows. Alternatively, the milk proportion of <16-C fatty acids has been shown to be greater in Jerseys than Holsteins (White et al., 2001; Sears et al., 2020), which may account for the breed difference in milk fat concentration considering that medium- and short-chain fatty acids originate from de novo synthesis in mammary tissues via ruminal precursors such as acetate and butyrate. Note that NDFD (present study; Aikman et al., 2008; Beecher et al., 2014; Olijhoek et al., 2018) and the ruminal molar proportion of acetate (Olijhoek et al., 2018, 2022), produced mostly by cellulolytic bacteria, were greater in Jerseys than Holsteins. Furthermore, milk lactose concentration was lower in Jersey versus Holstein cows, with this response not explained by a dilution effect caused by decreased milk volume in Jerseys. Olijhoek et al. (2018, 2022) reported that the ruminal molar proportion of propionate was lower in Jerseys compared with Holsteins in diets containing different F:C ratios. Ruminal propionate is the major gluconeogenic precursor in ruminants, potentially explaining the reduction in the concentration and yield of milk lactose in Jersey compared with Holstein cows.

Feed Efficiency and Milk N Efficiency

Our results showed that although FE_{MY} was lower in Jersey than Holstein cows, FE_{ECM} was not affected by breed. Uddin et al. (2020) also observed lower FE_{MY} in Jersey compared with Holstein cows fed diets in which alfalfa silage NDF partially replaced corn silage NDF at 2 levels of forage NDF. However, breed did not affect FE when it was calculated as the ratio between fat and protein-corrected MY over DMI in the study of Uddin et al. (2020). In contrast, Olijhoek et al. (2022) showed that FE_{ECM} was greater in Jerseys versus Holsteins in diets with concentrate levels ranging from 49% to 91% of the diet DM. In a subsequent experiment from the same laboratory, Børsting et al. (2023) reported a diet by breed interaction for FE_{ECM}, which decreased linearly with increasing concentrate levels (DM basis) from 49% to 91% only in Holsteins. Børsting et al. (2023) also observed diet by breed interactions for the concentration and yield of milk fat. Whereas both variables were lower in the greatest than the lowest concentrate level in the diet DM in Holsteins, they were similar in Jerseys (Børsting et al., 2023). Results from Olijhoek et al. (2022) and Børsting et al. (2023) demonstrated that milk fat was more stable in Jerseys than Holsteins in response to incremental dietary levels of concentrate, helping to explain the similar FE_{ECM} between breeds in the present meta-analysis. As discussed earlier, the ruminal molar proportion of acetate has been shown to be greater in Jersey than Holstein cows (Olijhoek et al., 2018, 2022), with acetate decreasing when the high concentrate diet was fed to Holsteins rather than Jerseys (Olijhoek et al., 2018). According to Olijhoek et al. (2022), the rumen of Jersey cows appeared to be less affected by elevated concentrate supply than that of Holstein cows, possibly because Jerseys have a better capacity to buffer organic acids due to more intense chewing per kilogram of DMI compared with Holsteins. However, research investigating the interactions among dietary-induced ruminal acidosis, ruminal biohydrogenation of fatty acids, and milk fat synthesis comparing these 2 breeds is lacking.

There was no effect of breed on MNE, corroborating previous studies in which Holstein and Jersey cows received diets containing different dietary levels of CP and NDF (Kauffman and St-Pierre, 2001; Uddin et al., 2020), F:C ratios (Børsting et al., 2023), and silage sources (Uddin et al., 2020). Although our MNE and FE_{ECM} results indicate that Holstein and Jersey cows have comparable efficiency to convert nutrients into milk protein and ECM, Olthof et al. (2023) showed that Holsteins were more profitable than Jerseys using data from 3 commercial dairy herds in Michigan. Both Kauffman and St-Pierre (2001) and Uddin et al. (2020) reported no effect of breed on fecal and urinary N excretion and retained N, all expressed as a proportion of N intake, further showing similar N utilization when comparing Holstein and Jersey cows. However, the output (g/d) of fecal and urinary N was greater in Holstein than Jersey cows (Kauffman and St-Pierre, 2001; Uddin et al., 2020), implying more environmental N pollution if raising Holsteins, assuming similar herd structures and feeding and manure management. In support, Uddin et al. (2021) showed, through a life-cycle assessment, that GHG emissions from manure management were 30% greater in Holsteins than Jerseys (0.52 vs. 0.40 kg of CO_2 Eq/kg of fat and protein-corrected milk) in diets containing different levels of forage NDF and silage sources.

Meta-Regression and Subgroup Analyses

We conducted a meta-regression to identify potential sources of heterogeneity and assess whether the continuous covariates year of publication, DIM, and dietary concentrations (DM basis) of NDF, CP, and forage affected the variables evaluated in the meta-analysis. In addition, a separate meta-regression examined the effect of the categorical covariate forage type on different variables using corn silage as the reference group (i.e., the intercept in the meta-regression equations). Altogether, the continuous (except BW, yield and concentration of milk lactose, milk true protein concentration, and ECM yield [% of BW and % of MBW]) and categorical covariates (except DMI [% of BW] and ECM yield [% of BW]) explained, based on the adjusted R^2, <50% of the heterogeneity of most observed and estimated variables (apart from unknown factors not included in the meta-regression that possibly affected the WMD estimates).

Year of publication narrowed the WMD of DMI (kg/d), milk true protein yield, and N intake, and widened that of milk fat and milk true protein concentrations and ECM yield (% of MBW), favoring Jerseys over Holsteins with each incremental increase in publication year. In contrast, year of publication widened the WMD of milk lactose concentration, benefiting Holstein versus Jersey cows. It can be hypothesized that the more recent publications used in our meta-analysis included Jerseys with improved genetic merit, which together with changes in management practices, collectively interacted to narrow or widen the differences between breeds for important production and economic variables favoring the Jersey breed. In fact, our subgroup analyses showed that, particularly in studies published >2008, breed differences either narrowed (DMI [kg/d] and milk true protein yield) or widened (concentrations of milk fat and milk true protein and ECM yield [% of MBW]) benefiting Jerseys at the expense of Holsteins.

Days in milk either narrowed (MY, milk fat yield, ECM yield [kg/d], and milk N yield) or widened (milk fat concentration, DMI [% BW and % of MBW], and ECM yield [% of BW and % of MBW]) the WMD of these variables, favoring Jersey over Holstein cows with each incremental increase in DIM. In contrast, the breed difference for BW widened in response to DIM, favoring Holsteins over Jerseys. Greater DMI (% of BW and % of MBW) in Jersey versus Holstein cows generally explains the narrowed (MY, milk fat yield, ECM yield [kg/d], and milk N yield) or widened (milk fat concentration and ECM yield [% of BW and % of MBW]) breed differences with the advance in DIM, benefiting the Jersey breed. Our subgroup analyses also revealed that, especially at >100 DIM, the differences between breeds either widened (milk fat concentration, DMI [% of MBW], and ECM yield [% of BW and % of MBW]) or narrowed (milk fat yield and ECM yield [kg/d]) in favor of Jerseys over Holsteins. This further reinforces the hypothesis that Jerseys appeared to have closed the gap in milk fat yield and ECM yield (kg/d) relative to Holsteins because the WMD of DMI (% of MBW) was greater in Jersey than Holstein cows at >100 DIM, but did not differ between breeds at ≤100 DIM. Alternatively, the breed difference in BW, favoring Holsteins as shown by our subgroup analysis, was wider at >100 DIM than at ≤100 DIM, suggesting that Jerseys may rely more heavily on body reserves to keep up with the lactation demands past 100 DIM or that they may deposit less body reserves as the lactation progresses compared with Holstein cows. However, previous research revealed that Jerseys had greater BCS than Holsteins over the course of the lactation (Washburn et al., 2002; Prendiville et al., 2009). Therefore, it is unclear why the breed differences in BW widened at >100 DIM, particularly considering the fact that the WMD of DMI (% of MBW) also widened at >100 DIM, but favoring Jerseys versus Holsteins.

Dietary concentration of NDF affected the WMD of BW (tendency), MY, and FE_{MY}, indicating that Jersey cows narrowed the gap compared with Holstein cows for these variables with each percentage-unit increase of NDF in the diet DM. The narrowed difference in MY between breeds with the increase in the NDF content of the diet suggests that Jerseys seem to be more effective than Holsteins at using energy from fibrous sources to produce milk, ultimately supporting the Jersey breed to reduce the gap in FE_{MY} relative to Holsteins. However, the difference between breeds narrowed (ECM yield [% of BW]) or widened (milk lactose yield) in favor of Holsteins over Jerseys with the increase in dietary concentration of NDF. Why Holsteins narrowed the difference in ECM yield (% of BW) relative to Jerseys in response to increased dietary level of NDF is unclear, particularly because NDFD has been shown to be greater in Jersey than Holstein cows (Olijhoek et al., 2018, 2022). In fact, our results also showed that NDFD was greater in Jerseys versus Holsteins even though present NDFD data should be interpreted cautiously because of the low number of observations for breed comparison (n = 14).

Dietary concentration of CP affected the WMD of BW and milk lactose content, with the difference between breeds narrowing in favor of Jerseys over Holsteins with each percentage-unit increase of CP in the diet DM. The narrowed breed difference in BW with the increase in dietary CP level suggests that, compared with Holstein cows, Jerseys may partition more dietary N toward muscle mass. However, it is unclear why the WMD of milk lactose concentration narrowed in response to an incremental increase in dietary CP because lactose is generally not affected by CP intake (Olmos Colmenero and Broderick, 2006; Yang et al., 2022). We also found that the difference in N intake between breeds widened with each percentage unit increase of CP in the diet DM favoring Holsteins over Jerseys. This may be associated with Holsteins consuming more DM than Jerseys (WMD = −4.15 kg/d) even though DMI (kg/d, % of BW, and % of MBW) was not affected by dietary CP level. Despite increased N intake in Holsteins, the WMD of MNE was not affected by dietary CP concentration. Studies evaluating N utilization in Holstein versus Jersey cows fed different dietary levels of CP are limited. Kauffman and St-Pierre (2001) reported that whereas MNE did not differ, productive N (milk N + retained N; % of N intake) tended to be lower in Jerseys than Holsteins in diets with varying CP (13% vs. 17% CP) and NDF (30% vs. 40%) concentrations. As discussed earlier, breed-specific postabsorptive metabolism of AA and energy likely exist for Holstein and Jerseys cows, warranting further investigations.

Forage level affected the WMD of milk lactose yield, with the difference between breeds narrowing in favor of Jerseys over Holsteins with each percentage-unit increase of forage in the diet DM. However, previous research showed no effect of diets with different F:C ratios or diet by breed interactions on the concentration and yield of milk lactose when comparing Holstein versus Jersey cows (Olijhoek et al., 2018, 2022; Børsting et al., 2023). It is unclear why the difference between breeds for milk lactose yield narrowed benefiting Jerseys in response to increased forage level, particularly because milk lactose content and MY were not affected by the concentration of forage in the diet DM. In contrast, the WMD of milk true protein concentration and ECM yield (% of MBW) narrowed favoring Holsteins over Jerseys with the incremental increase of forage in the diet DM. Note that these results were not accompanied by an effect of forage level on DMI (kg/d, % of BW, and % of MBW), milk true protein yield, and ECM yield (kg/d and % of BW). Therefore, the forage level effect on both milk true protein concentration and ECM yield (% of MBW) should be interpreted cautiously.

Diets containing corn silage affected the WMD of most observed and estimated variables. Specifically, the difference between breeds favored Holsteins over Jerseys for DMI (kg/d), BW, MY, milk true protein yield, ECM yield (kg/d), FE_{MY}, N intake, and milk N yield, and benefited Jerseys over Holsteins for the concentrations of milk fat and milk true protein and DMI (% of BW and % of MBW). Compared with corn silage–based diets, feeding cows alfalfa/grass-clover mix widened the breed differences for DMI (kg/d), MY, yields of milk fat and milk true protein, ECM yield (kg/d), N intake, and milk N yield, favoring Holsteins over Jerseys. In addition, in comparison to corn silage, the WMD of DMI (% of MBW) narrowed between breeds in cows fed alfalfa/grass-clover mix, also benefiting Holsteins versus Jerseys. Contrarily, feeding alfalfa/grass-clover mix did not change the effect of corn silage on the concentrations of milk fat and milk true protein between breeds.

We are aware of a single publication (i.e., Uddin et al., 2020) in which diets containing predominantly NDF from corn silage or alfalfa silage were fed to Holstein and Jersey cows. In Uddin et al. (2020) study, authors detected breed by forage NDF source interactions only for DMI (kg/d and % of BW). Specifically, Uddin et al. (2020) showed that although the average DMI (kg/d) of Jerseys was 25.4% lower than that of Holsteins, this difference narrowed when corn silage (−28%) instead of alfalfa silage (−41%) was the main forage source in the diet DM. They further showed that, on average, DMI (% of

BW) was 10.2% greater in Jersey versus Holstein cows, with this difference widening with corn silage– versus alfalfa silage–based diets (+13% and +5%, respectively). Data from Uddin et al. (2020) indicate that compared with alfalfa silage, corn silage appears to be more effective for stimulating DMI (kg/d and % of BW) in Jerseys than Holsteins. In the present study, corn silage–based diets resulted in greater DMI (% of BW) in Jersey versus Holstein cows, but it was similar (i.e., not different from zero) across breeds in the subset of comparisons in which cows were fed alfalfa/grass-clover mix diets. In contrast, compared with corn silage, feeding alfalfa/grass-clover mix narrowed the WMD of DMI (% of MBW) between breeds, benefiting Holsteins over Jerseys. Therefore, collective effects of the forage types corn silage and alfalfa/grass-clover mix on DMI (% of BW and % of MBW) did not result in one breed having a clear advantage over the other. However, it is conceivable that breed differences in ruminal fill, passage rate, retention time, and NDF digestibility potentially played a role because these factors are known to affect DMI. For instance, Aikman et al. (2008) showed that estimated passage rate of particles out of the rumen was faster in Jerseys versus Holsteins, which was partially explained by their ability to reduce particle size more efficiently than Holsteins. Aikman et al. (2008) also observed that undigested feed took longer to pass through the gastrointestinal tract of Holsteins compared with that of Jerseys because of the longer mean retention time of particles within the rumen in the larger versus the smaller breed.

Compared with corn silage–based diets, the forage type others further widened the breed differences for the concentrations of milk fat and milk true protein, DMI (% of BW) and ECM yield (% of BW and % of MBW), all in favor of Jersey than Holstein cows. These results suggest that relative to Holsteins, Jerseys seem to benefit more from diets in which corn silage and alfalfa/grass-clover mix make up <55% of the total DMI and from a more diversified portfolio of forage sources.

CONCLUSIONS

Dry matter intake and ECM yield, expressed as proportions of BW and MBW, were greater in Jerseys versus Holsteins. Whereas FE_{MY} was lower in Jerseys than Holsteins, FE_{ECM} and MNE did not differ between breeds. Year of publication affected the breed differences for DMI (kg/d), milk true protein yield, concentration of milk fat and true protein, ECM yield (% of MBW), and N intake, favoring Jerseys over Holsteins. Days in milk also affected several variables, including DMI (% BW and % of MBW), MY, yields of milk fat and milk N, and ECM yield (kg/d, % of BW and % of MBW), benefiting Jerseys over Holsteins. Compared with corn silage, feeding alfalfa-grass/clover mix widened (DMI [kg/d], milk fat and milk true protein yields, and ECM yield [kg/d]) or narrowed (DMI, % of MBW) the breed difference in favor of Holsteins. However, relative to corn silage, the forage type others widened the breed differences for the concentrations of milk fat and true protein, DMI (% of BW), and ECM yield (% of BW and % of MBW), favoring Jerseys. Overall, FE_{ECM} and MNE were similar between breeds and the covariates year of publication, DIM, and forage type largely influenced the WMD of numerous variables.

NOTES

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Nonstandard abbreviations used: F:C ratio = forage:concentrate ratio; FE = feed efficiency; FE_{ECM} = feed efficiency calculated as ECM yield divided by DMI; FE_{MY} = feed efficiency calculated as milk yield divided by DMI; MBW = metabolic BW ($BW^{0.75}$); MNE = milk N efficiency; MY = milk yield; NDFD = apparent total-tract NDF digestibility; NS = nonsignificant; SED = standard error of the difference; WMD = weighted raw mean difference.

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